

[Review step by step / 逐项检查 →](#)

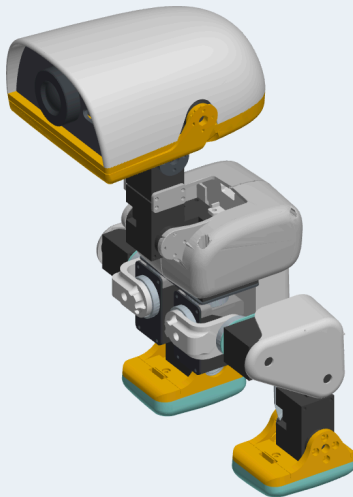
Microduck

Review the robot / 机器人评审

ENGINEERING REVIEW · 工程评审

Start with the robot, then open any part you want to examine. / 请先看整机，再选择要查看的零件。

Extract this folder once, then open START HERE.html in Chrome or Edge. No account, installation, server or internet connection is needed. On a phone, begin with START HERE.pdf. / 解压后用浏览器打开，无需安装或联网；手机请先查看PDF。



See the robot / 看整机

Rotate it, remove the covers and choose individual pieces. / 旋转整机、隐藏外壳、选择零件。

Look at each part / 看零件

One named folder per part, with STL, a 3D preview and available drawings. / 每个零件一个文件夹。

How it fits together / 看装配

Pictures explain the assembly stages. / 按图片查看装配阶段。

Electronics and wires / 看电子接线

Schematics, PCB files and wiring records. / 原理图、PCB与接线资料。

Shopping list / 看采购清单

Bought components and quantities. / 外购件与数量。

Questions to resolve / 看待确认事项

What the engineer must confirm before manufacturing. / 生产前由工程师确认。

All engineering files / 全部工程文件

Search every included file and see the source of each. / 搜索文件并查看来源。

About this review / 评审说明

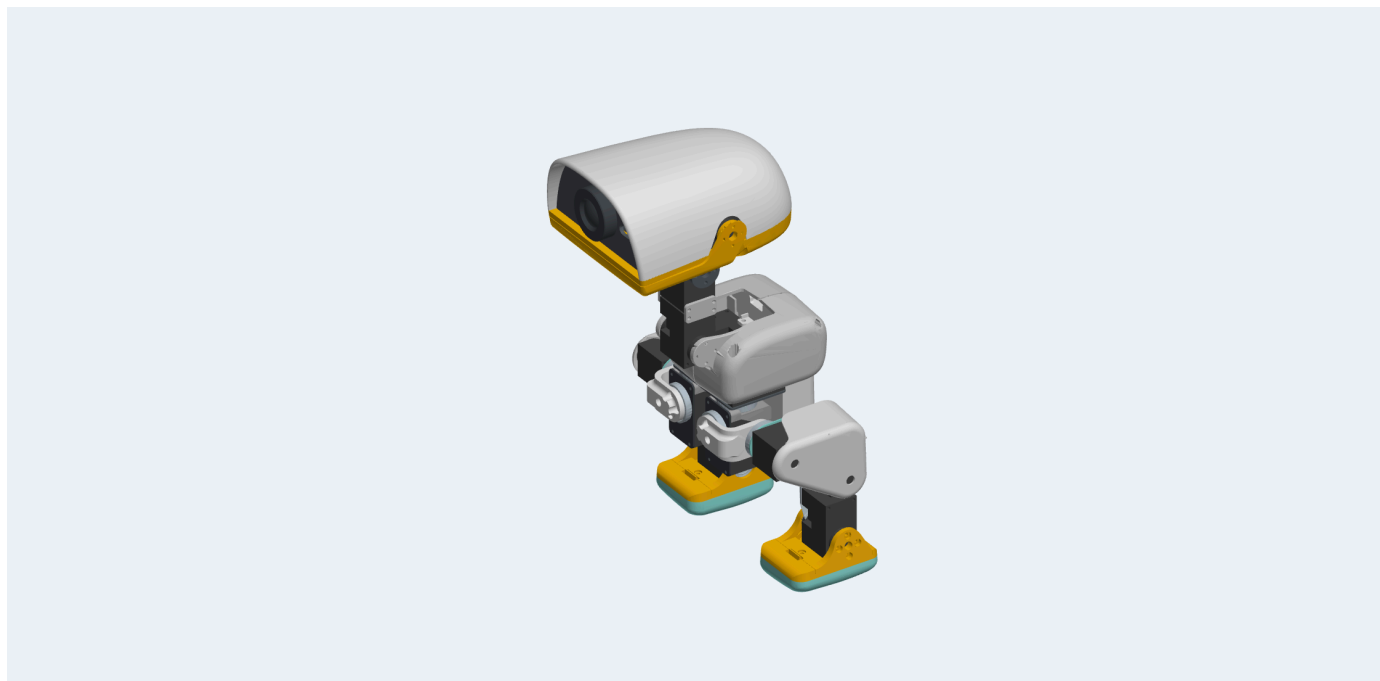
38 part types and 70 assembled pieces. Please review the design, connections and component choices. This package is not approved for manufacturing.

包含38种零件、70个装配件，供设计、接线与选型评审，尚未批准生产。

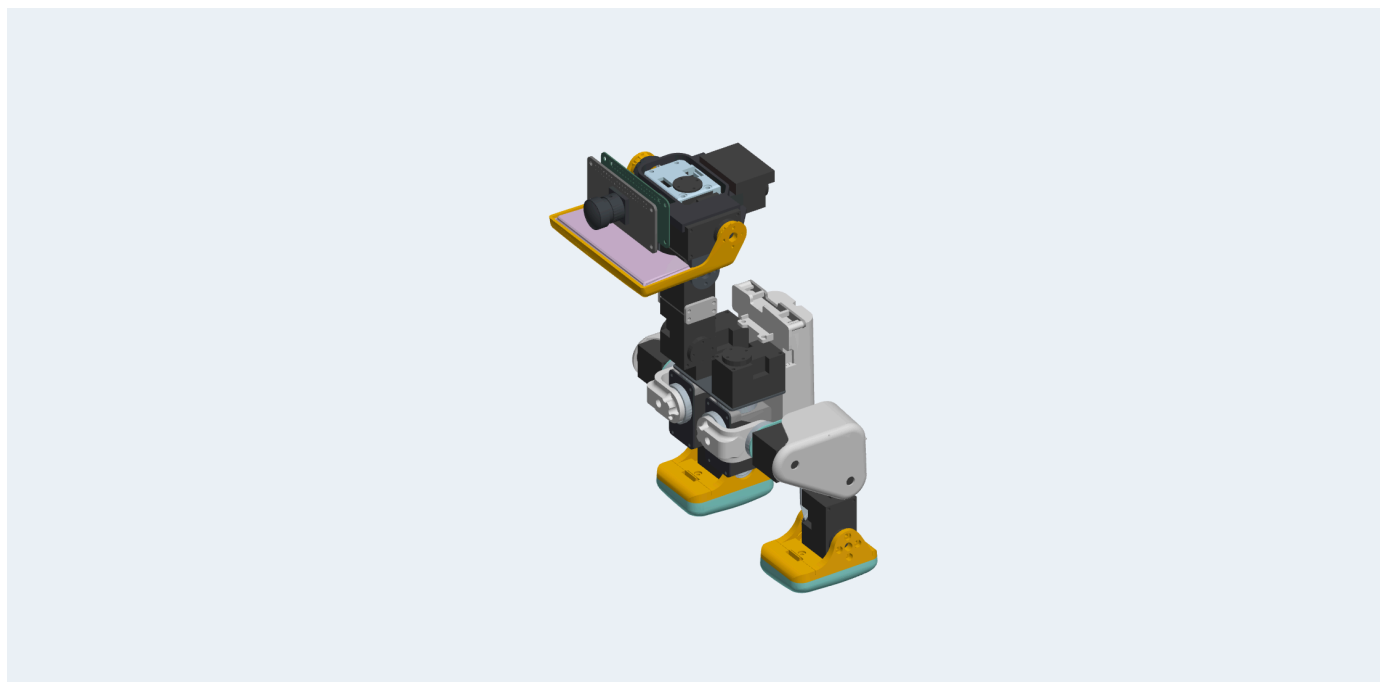
See the actual model / 查看实际模型

Rotate the model with one click, or review these pictures directly. / 点击进入3D，也可直接查看下列图片。

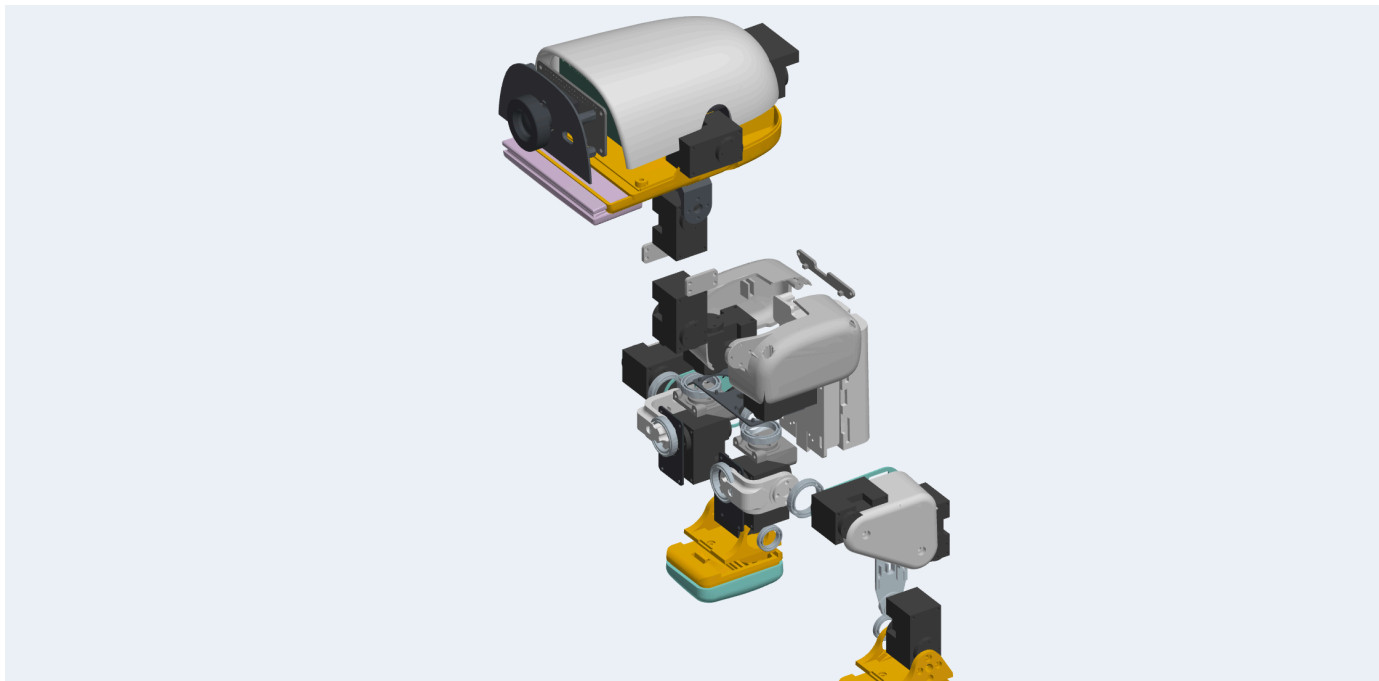
Whole robot / 整机



Covers removed / 隐藏外壳



Pieces separated for viewing / 分开查看零件

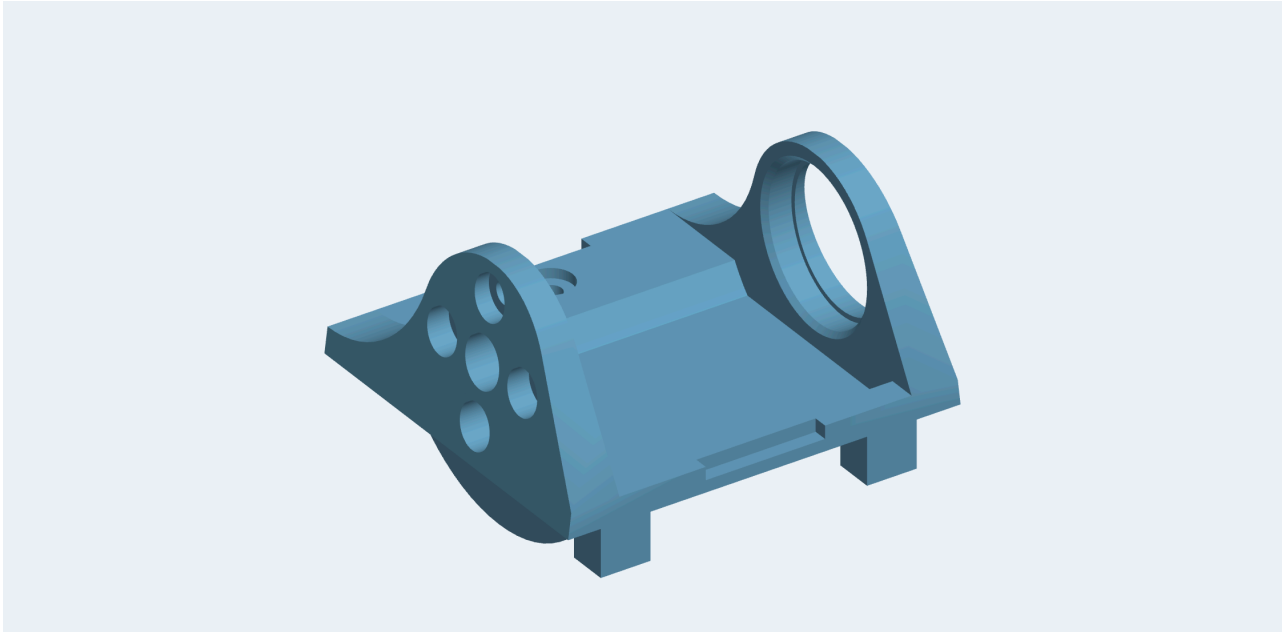


[Open interactive 3D / 打开交互式3D](#)

Every named print-list part / 打印清单零件总览

Click a picture to turn the actual STL around. Source labels distinguish reference meshes from reconstructed parts. / 点击图片旋转查看STL，来源标签区分参考网格和重建零件。

1 · Ankle Left

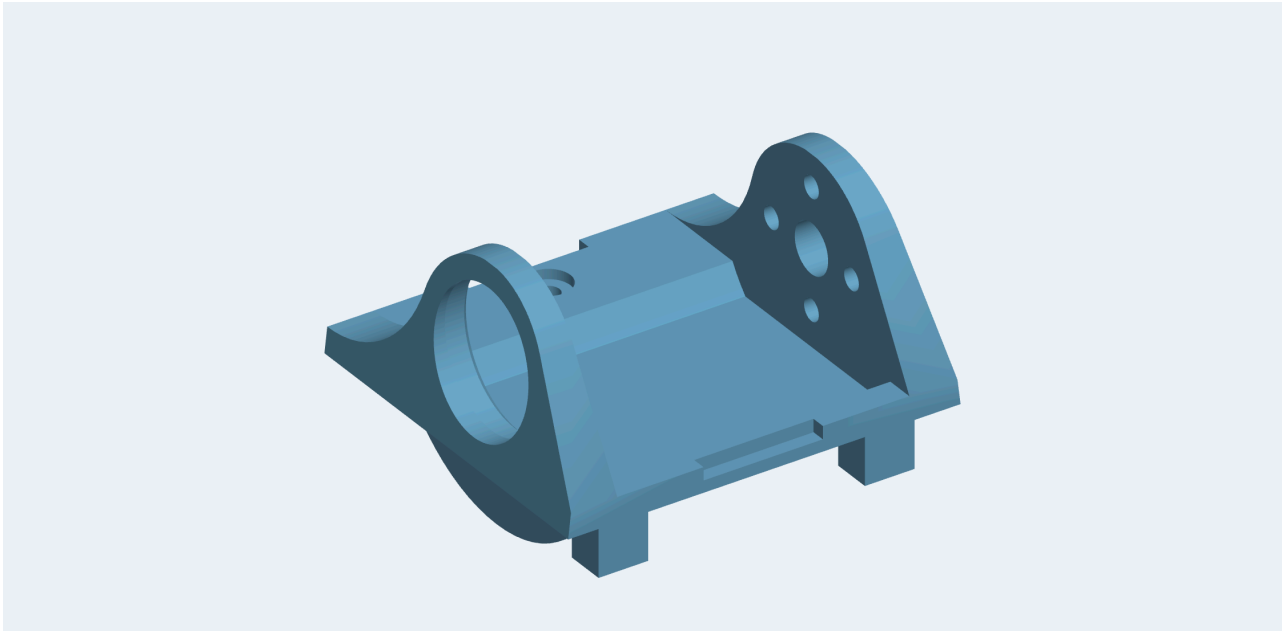


Quantity / 数量: 1

Reference geometry / 参考几何

[STL](#) · [3D view](#) / [查看3D](#)

2 · Ankle Right

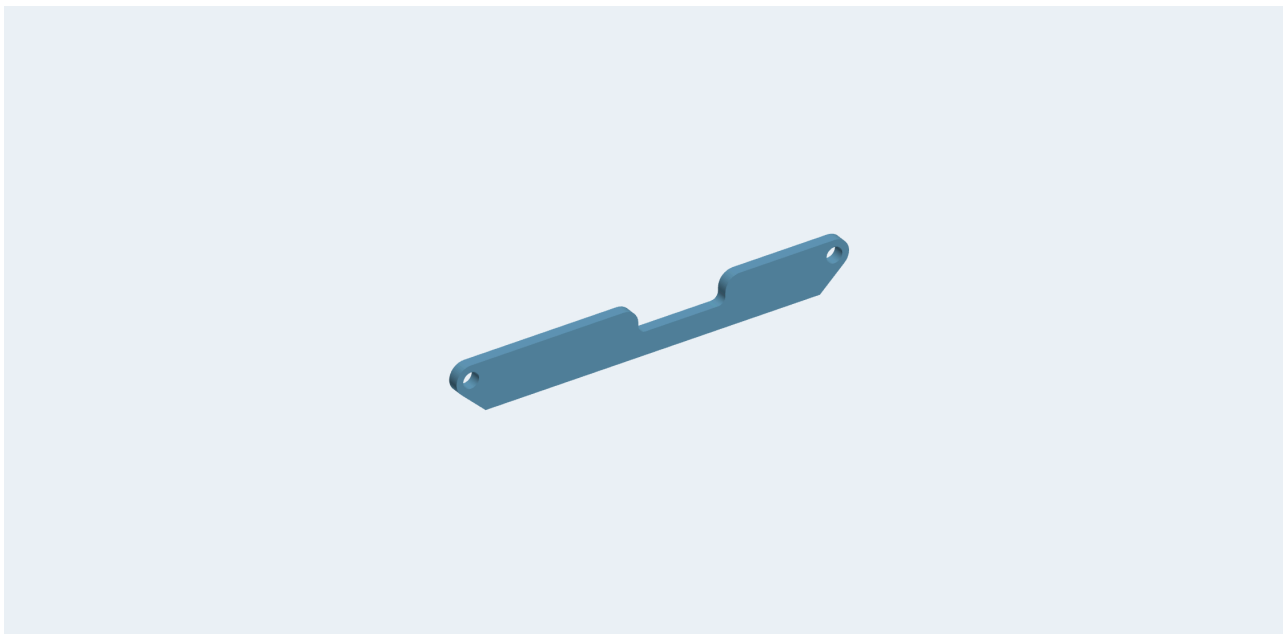


Quantity / 数量: 1

Reference geometry / 参考几何

[STL · 3D view / 查看3D](#)

3 · Banana Pcb Locker

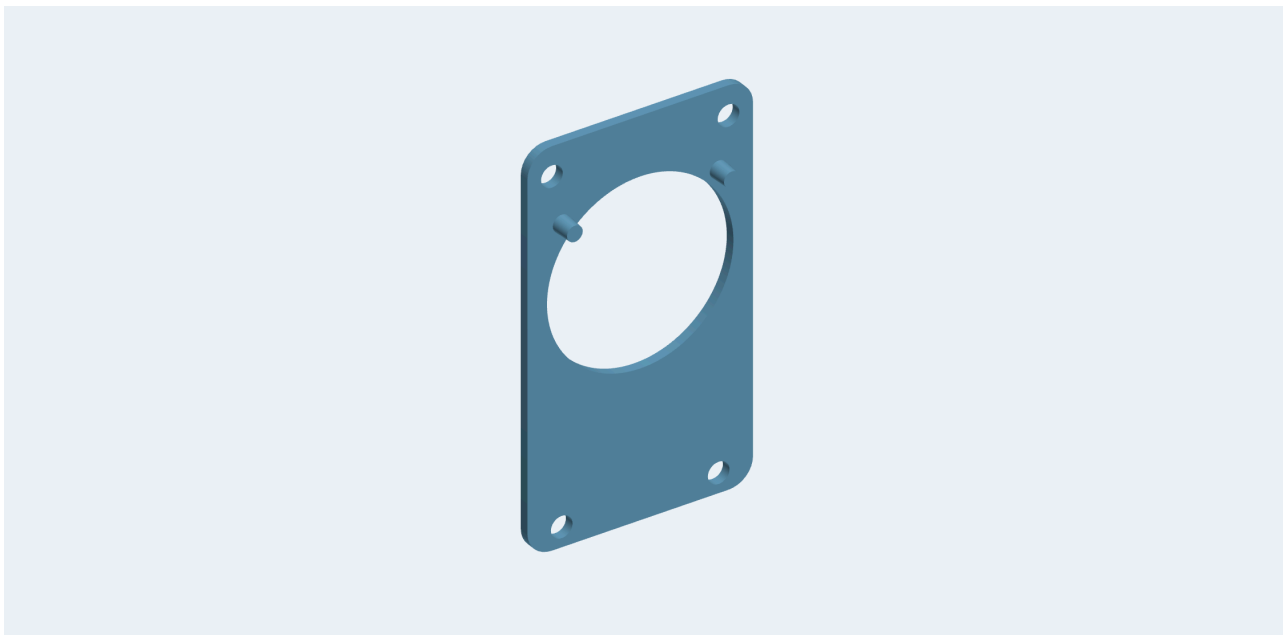


Quantity / 数量: 1

Reconstructed part / 重建零件

[STL · 3D view / 查看3D](#)

4 · Bearing Roll

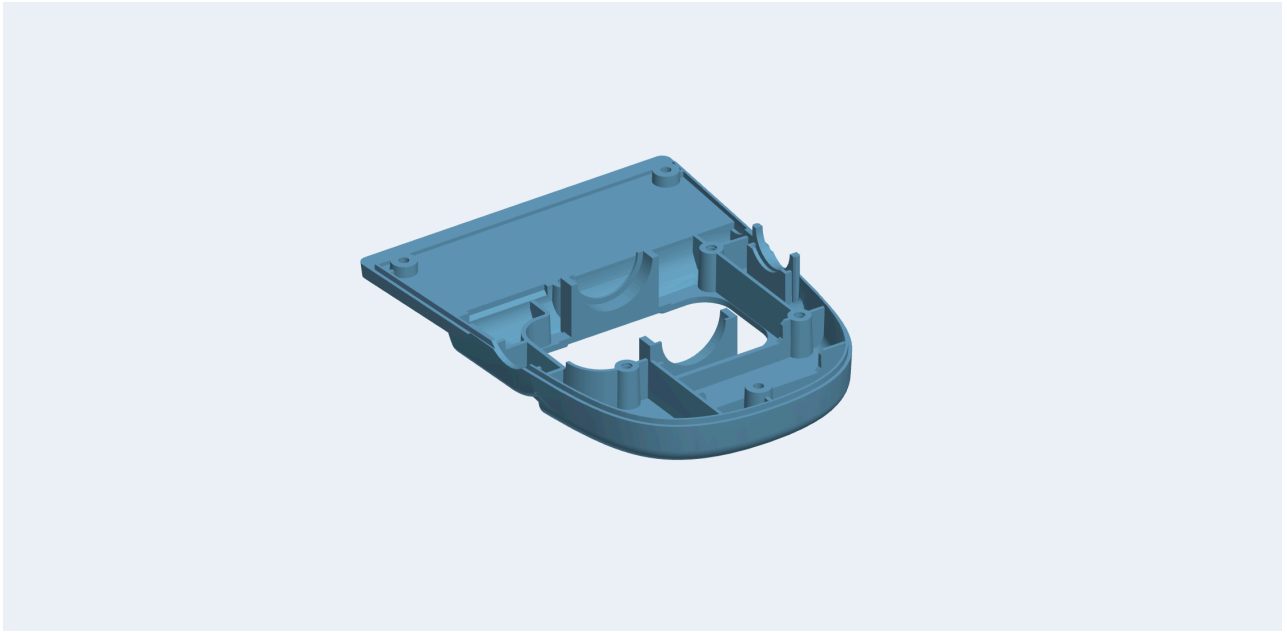


Quantity / 数量: 2

Reconstructed part / 重建零件

[STL · 3D view / 查看3D](#)

5 · Bottom Head Shell

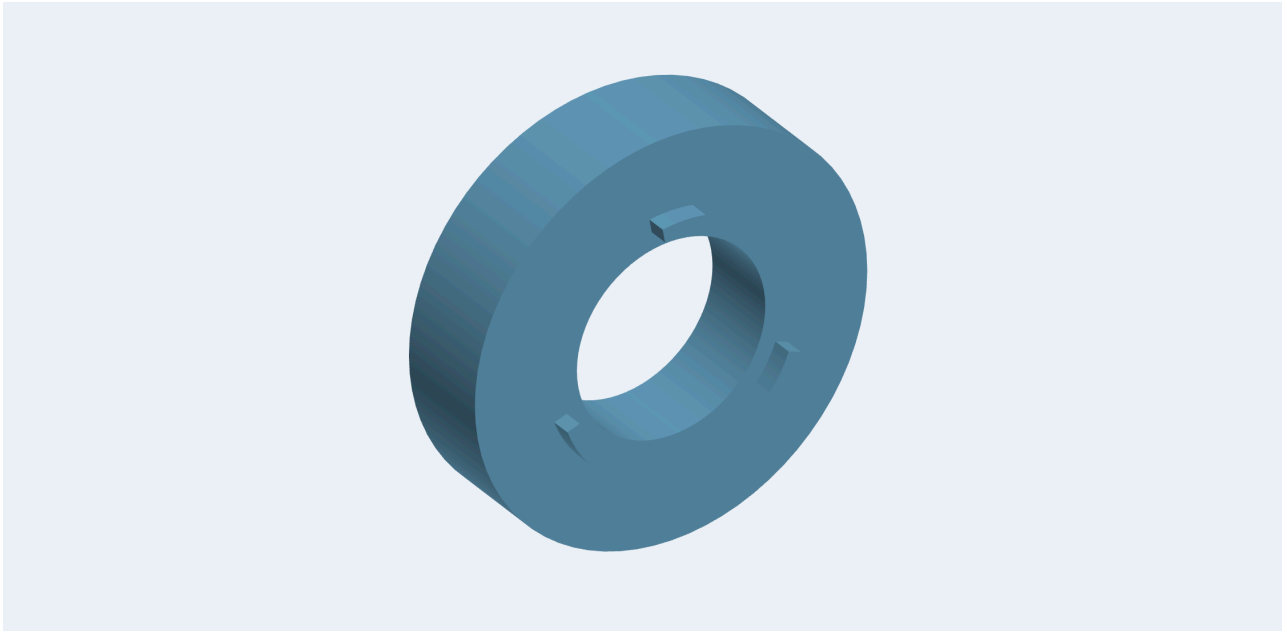


Quantity / 数量: 1

Reference geometry / 参考几何

[STL · 3D view / 查看3D](#)

6 · Eye Ring



Quantity / 数量: 1

Reference geometry / 参考几何

[STL · 3D view / 查看3D](#)

7 · Face Part

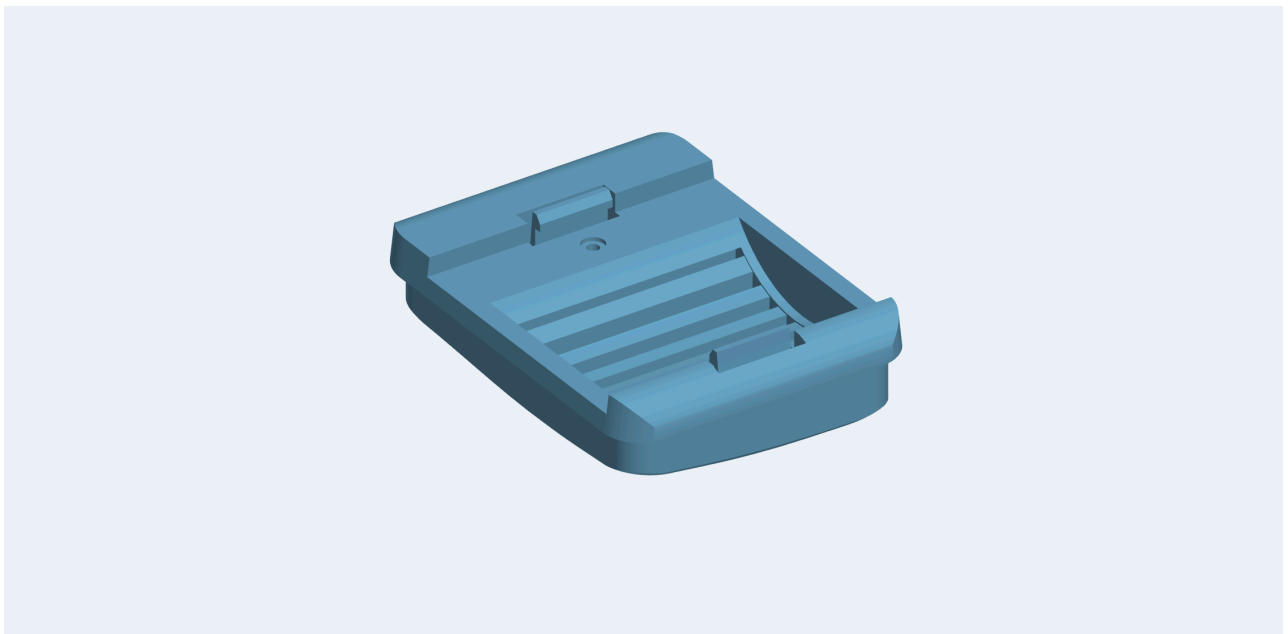


Quantity / 数量: 1

Reference geometry / 参考几何

[STL · 3D view / 查看3D](#)

8 · Foot Left

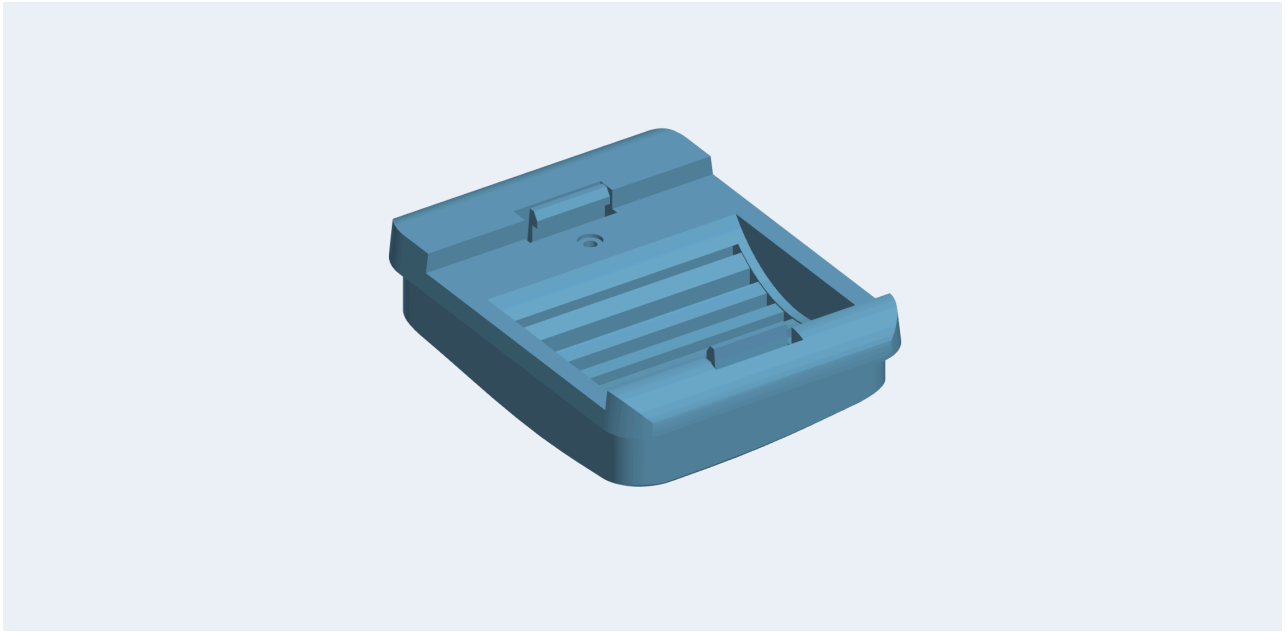


Quantity / 数量: 1

Reference geometry / 参考几何

[STL · 3D view / 查看3D](#)

9 · Foot Right

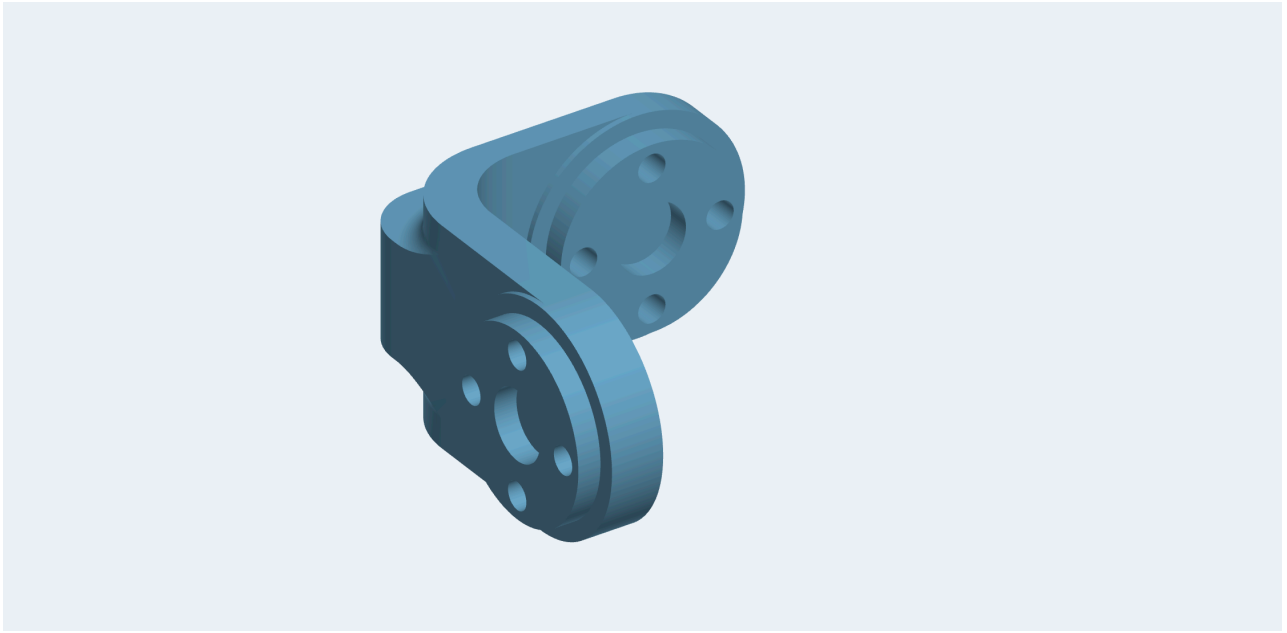


Quantity / 数量: 1

Reference geometry / 参考几何

[STL · 3D view / 查看3D](#)

10 · Hip Bracket

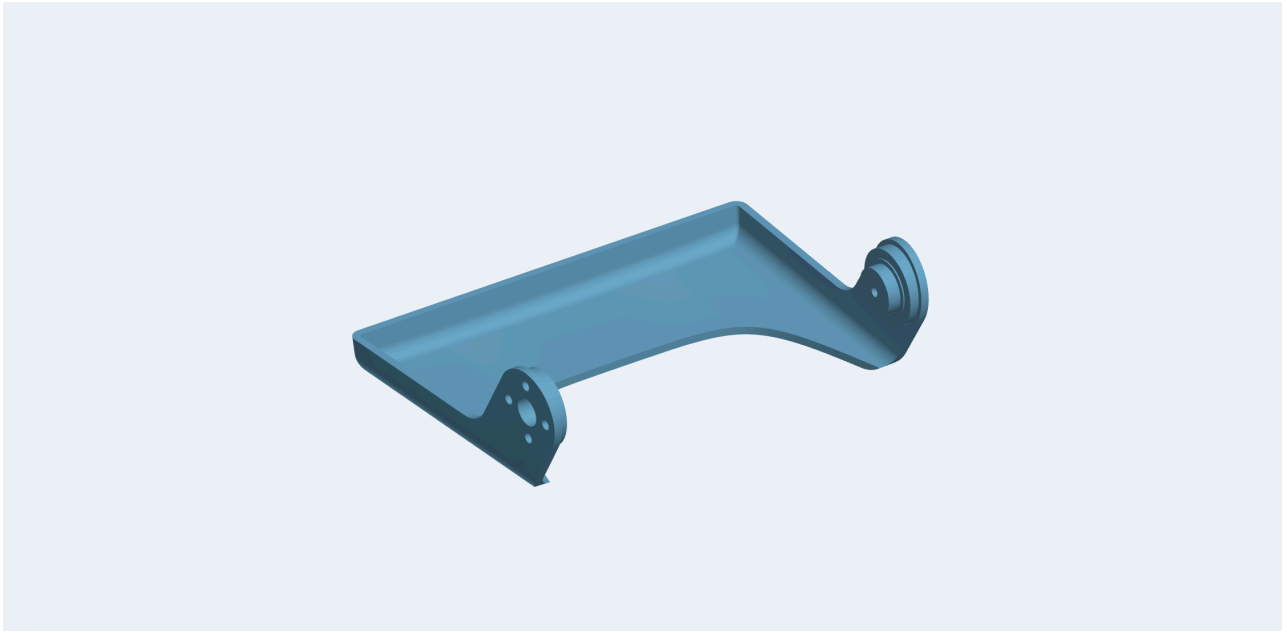


Quantity / 数量: 2

Current design / 当前设计

[STL](#) · [3D view](#) / [查看3D](#)

11 · Jaw

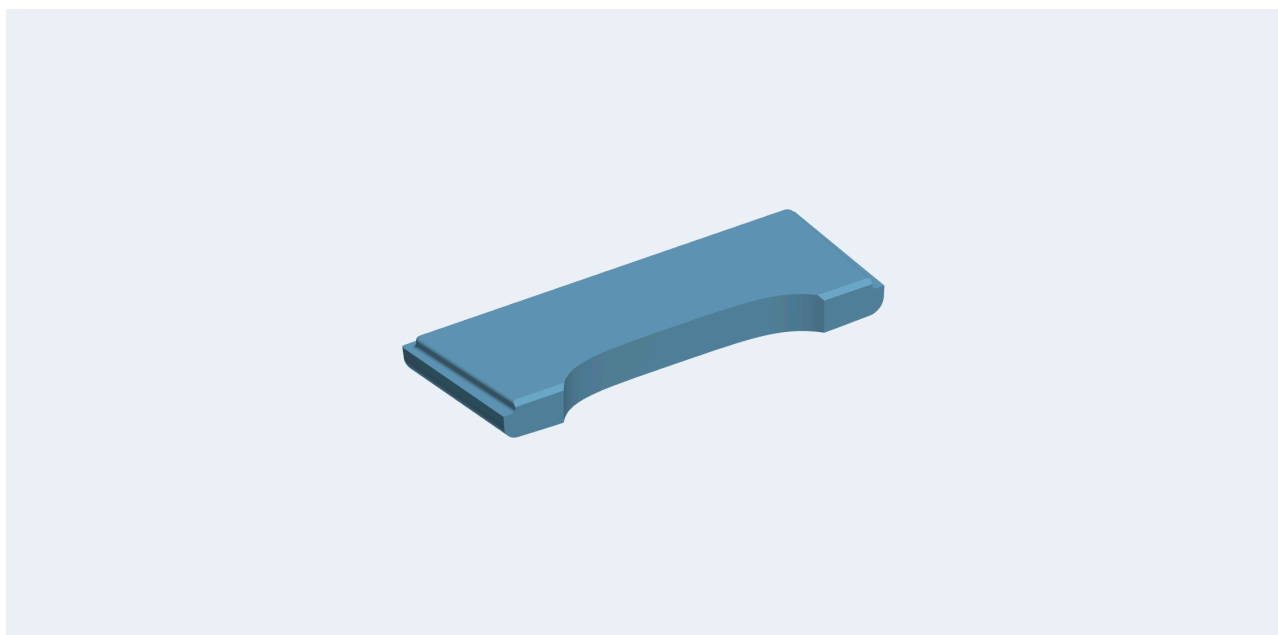


Quantity / 数量: 1

Reference geometry / 参考几何

[STL](#) · [3D view](#) / [查看3D](#)

12 · Jaw Soft

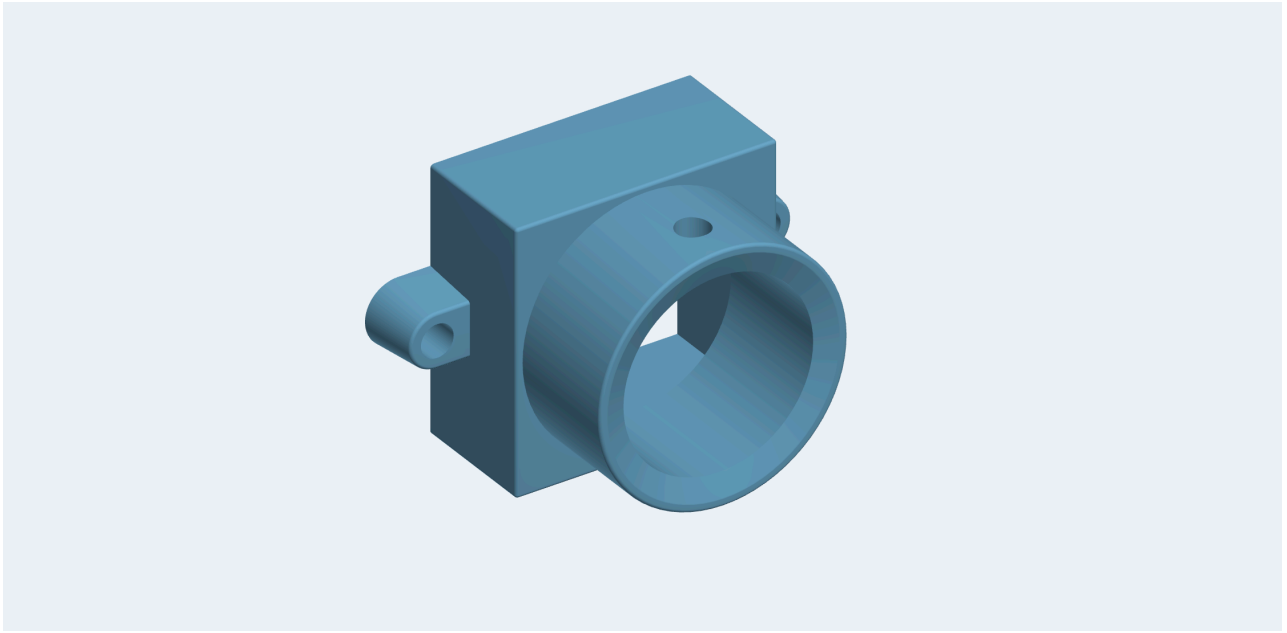


Quantity / 数量: 1

Reference geometry / 参考几何

[STL](#) · [3D view](#) / [查看3D](#)

13 · M12 Lens Holder

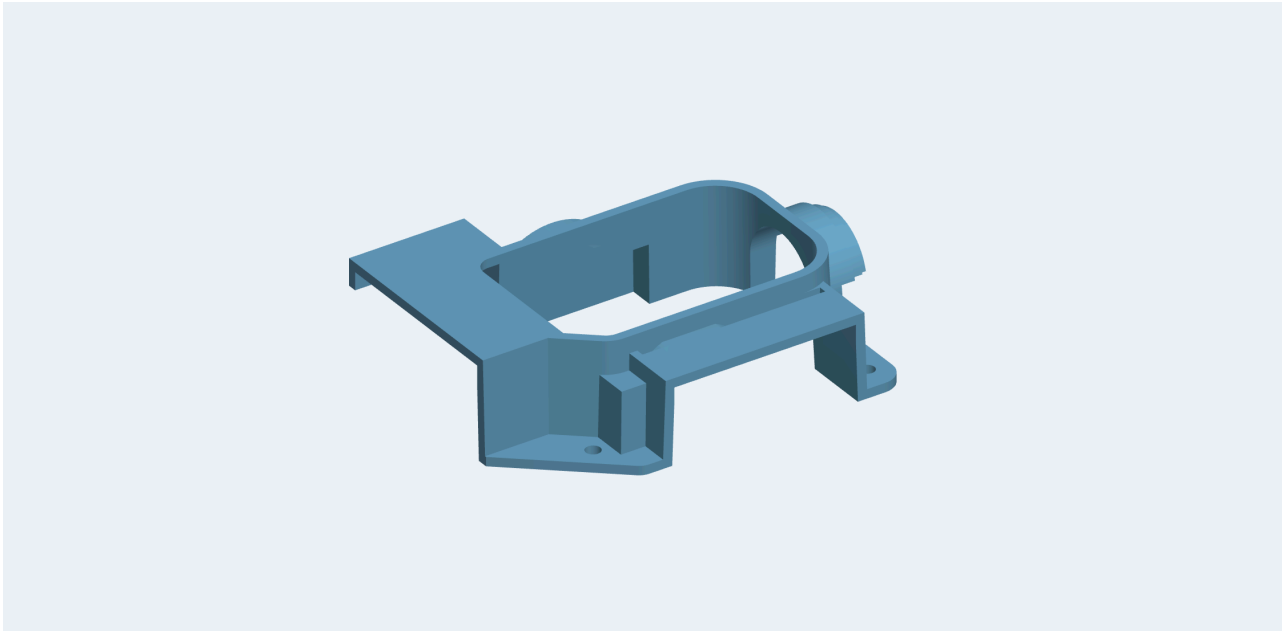


Quantity / 数量: 1

Reference geometry / 参考几何

[STL](#) · [3D view](#) / [查看3D](#)

14 · Motor Support



Quantity / 数量: 1

Reference geometry / 参考几何

[STL · 3D view / 查看3D](#)

15 · Neck Pitch Bracket

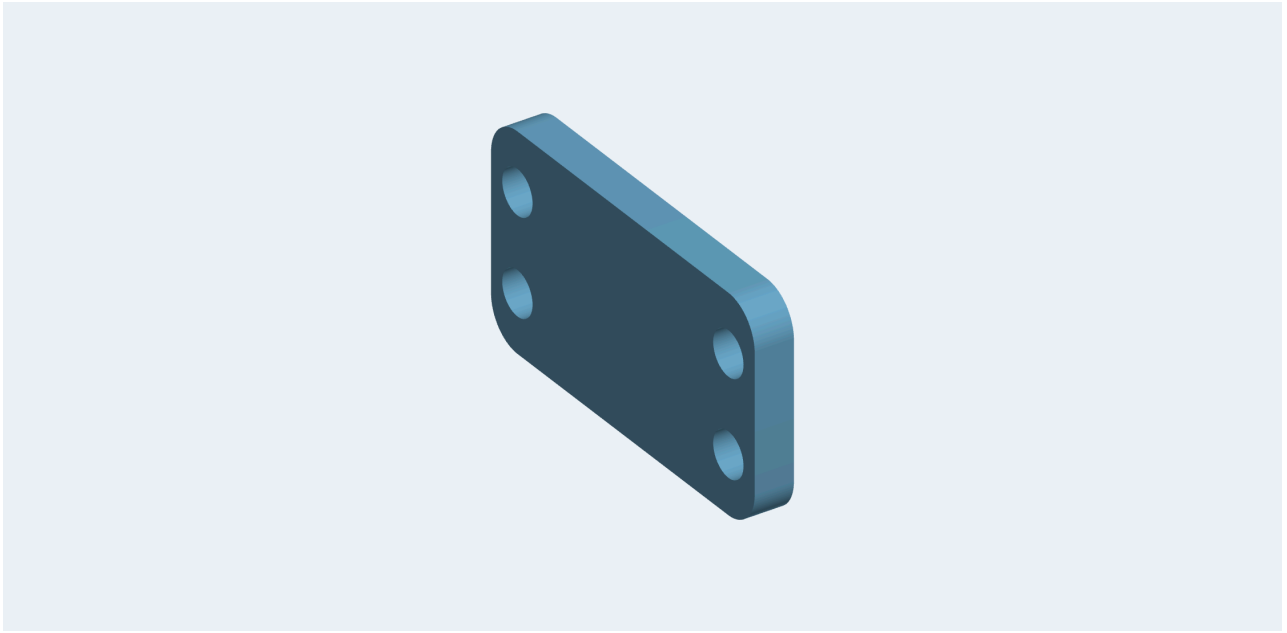


Quantity / 数量: 1

Reference geometry / 参考几何

[STL · 3D view / 查看3D](#)

16 · Neck Plate

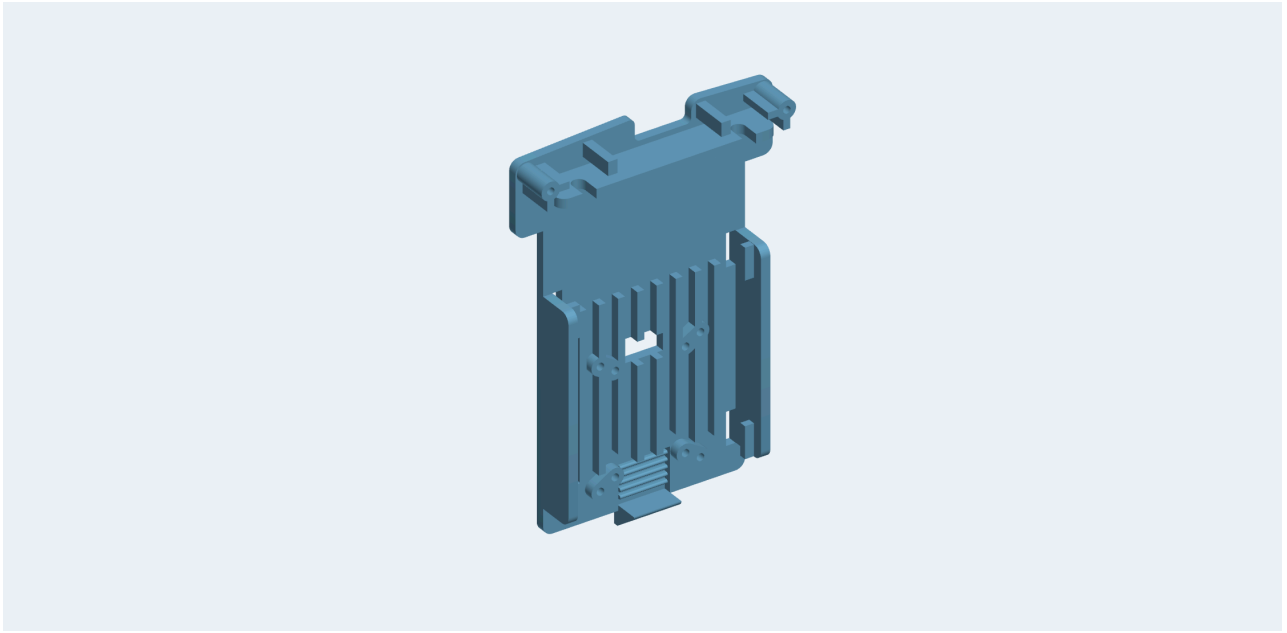


Quantity / 数量: 2

Reference geometry / 参考几何

[STL · 3D view / 查看3D](#)

17 · Power Support

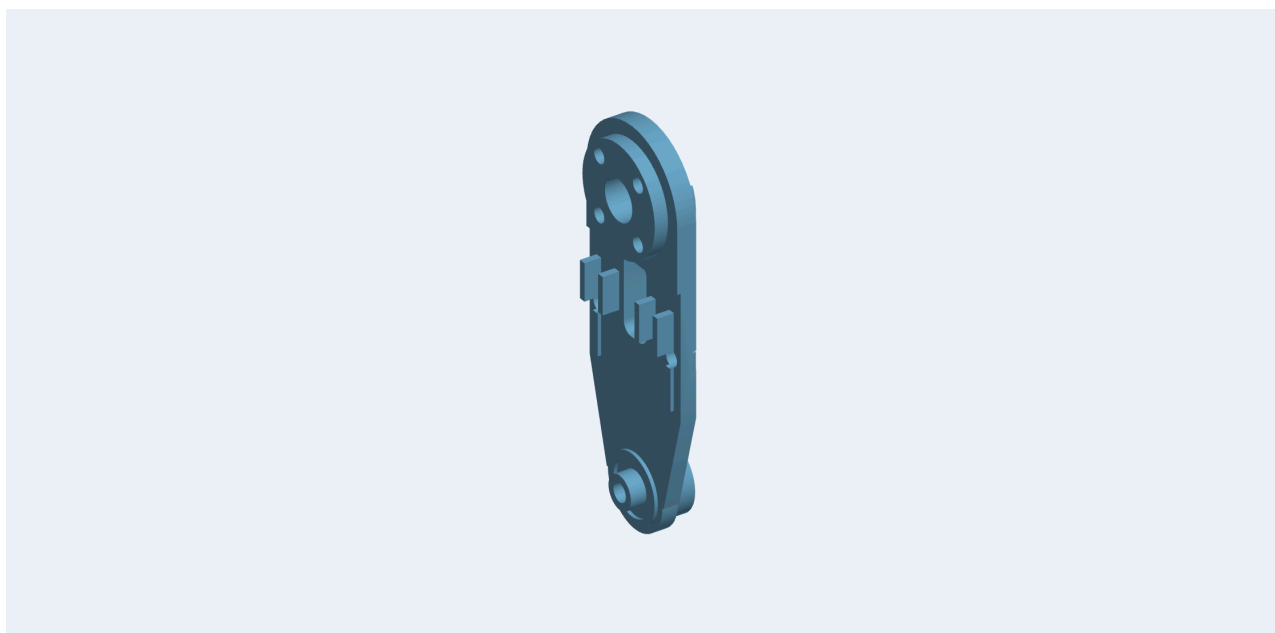


Quantity / 数量: 1

Current design / 当前设计

[STL · 3D view / 查看3D](#)

18 · Shin

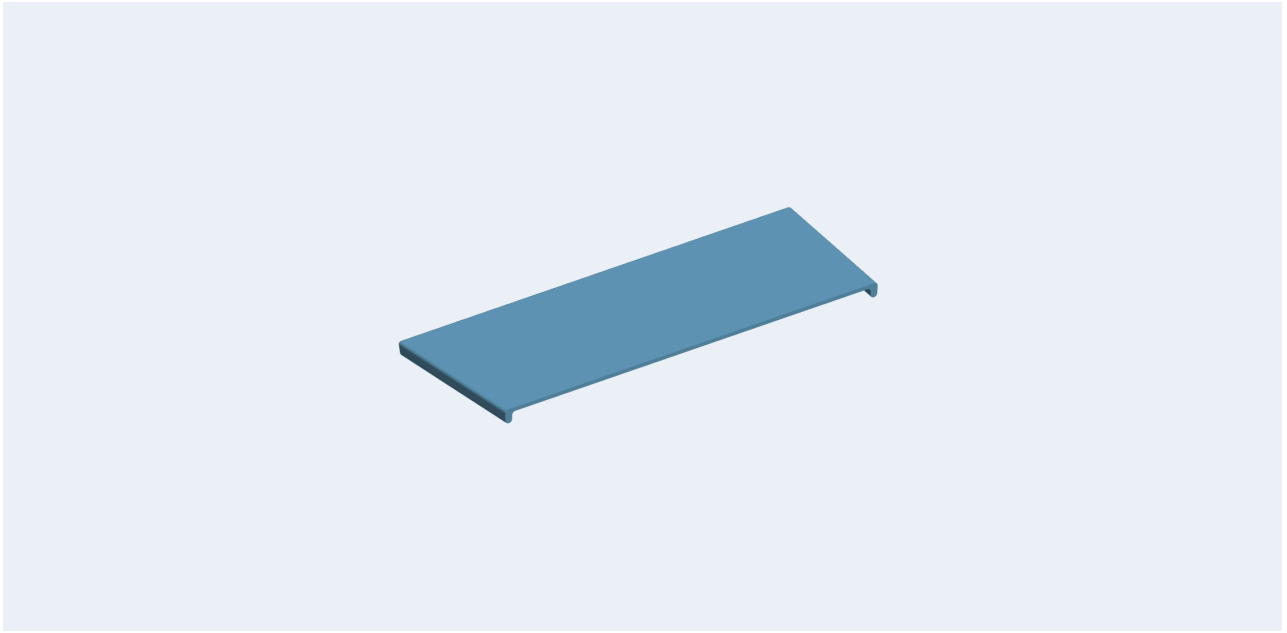


Quantity / 数量: 2

Reconstructed part / 重建零件

[STL](#) · [3D view](#) / [查看3D](#)

19 · Soft Mouth Top



Quantity / 数量: 1

Reference geometry / 参考几何

[STL](#) · [3D view](#) / [查看3D](#)

20 · Sole Left



Quantity / 数量: 1

Reference geometry / 参考几何

[STL](#) · [3D view](#) / [查看3D](#)

21 · Sole Right

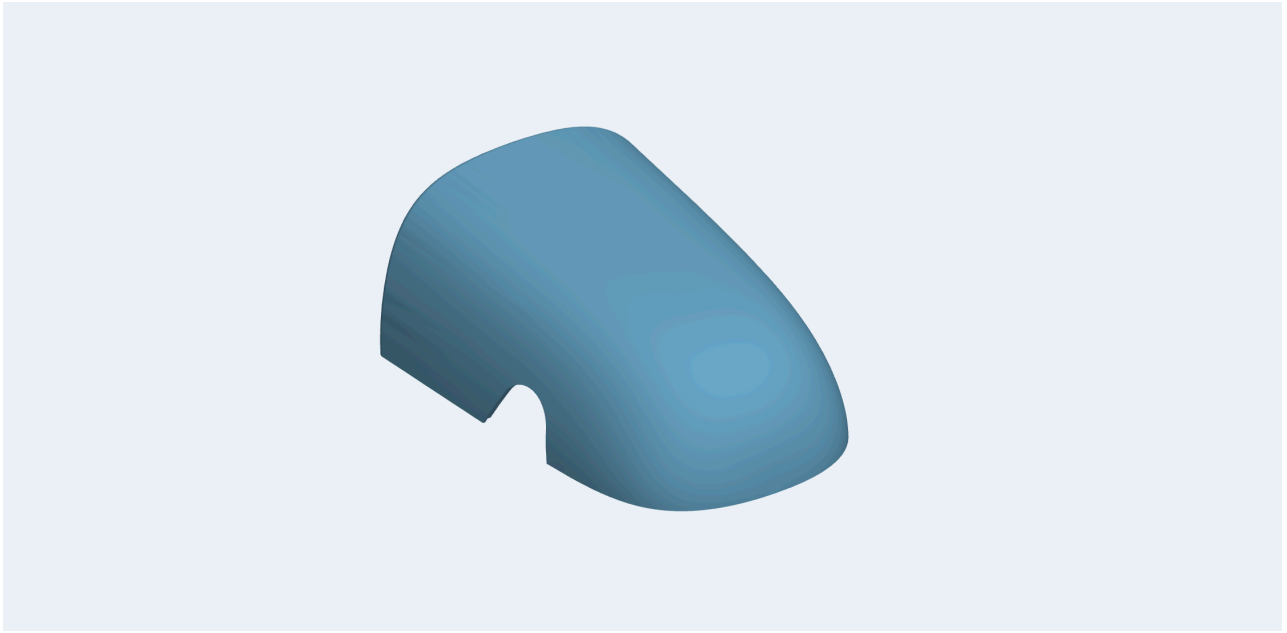


Quantity / 数量: 1

Reference geometry / 参考几何

[STL](#) · [3D view](#) / [查看3D](#)

22 · Top Head Shell

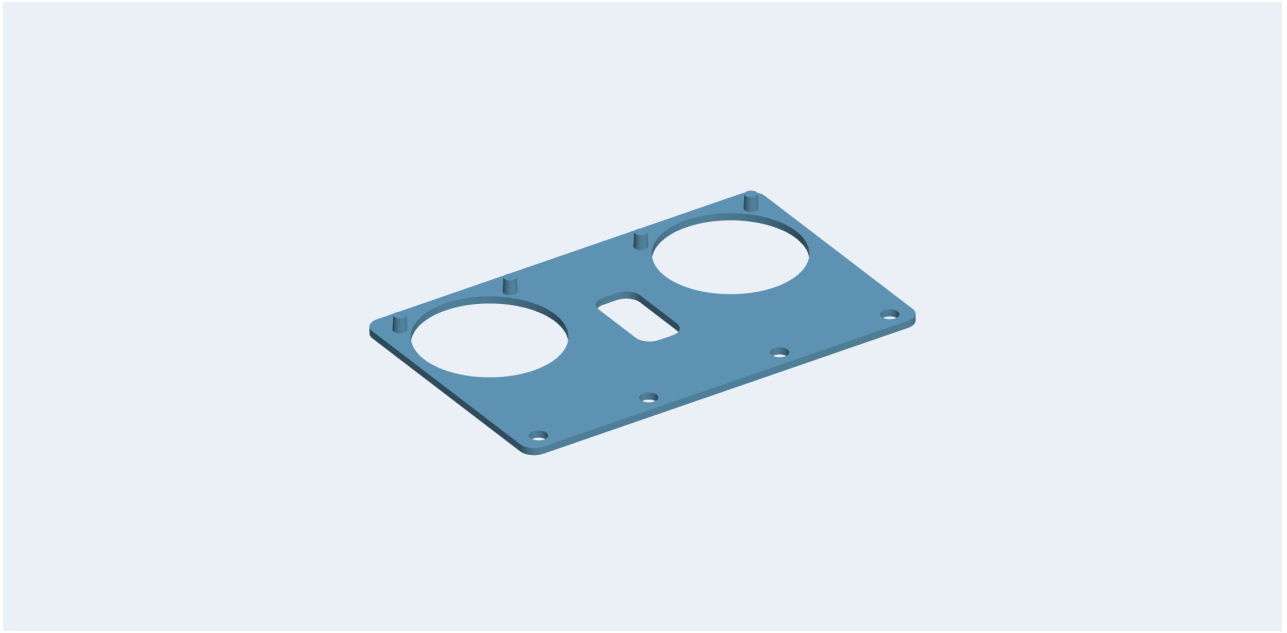


Quantity / 数量: 1

Reference geometry / 参考几何

[STL · 3D view / 查看3D](#)

23 · Trunk Base

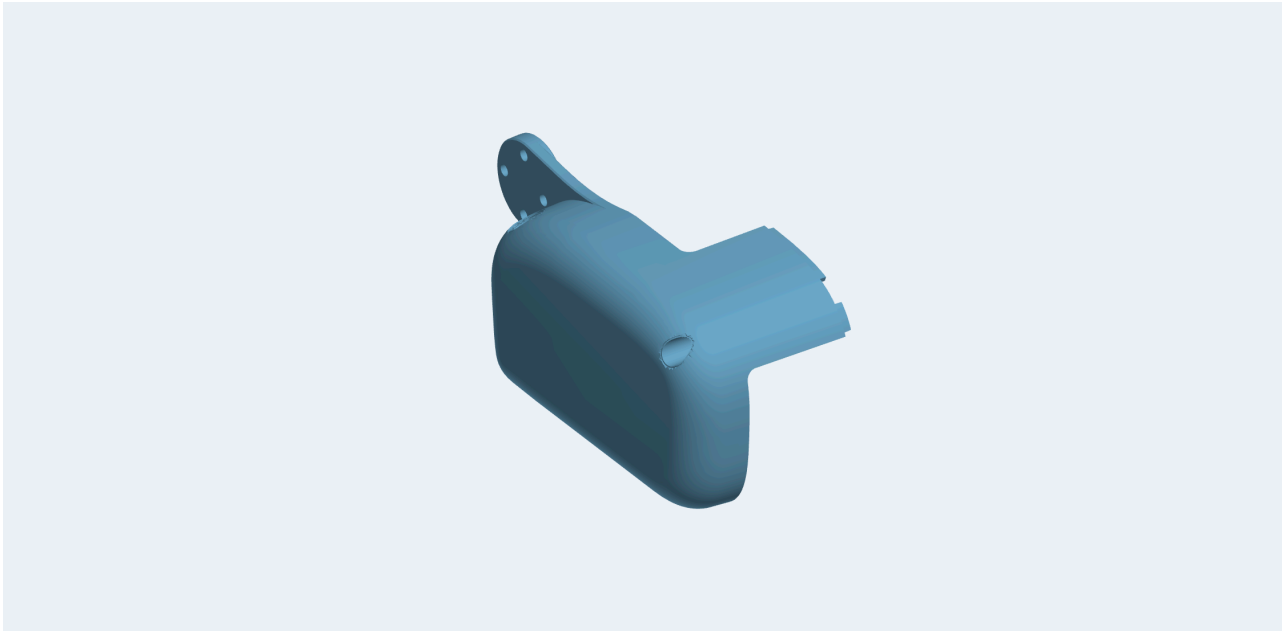


Quantity / 数量: 1

Reconstructed part / 重建零件

[STL · 3D view / 查看3D](#)

24 · Trunk Shell Left



Quantity / 数量: 1

Reference geometry / 参考几何

[STL · 3D view / 查看3D](#)

25 · Trunk Shell Right

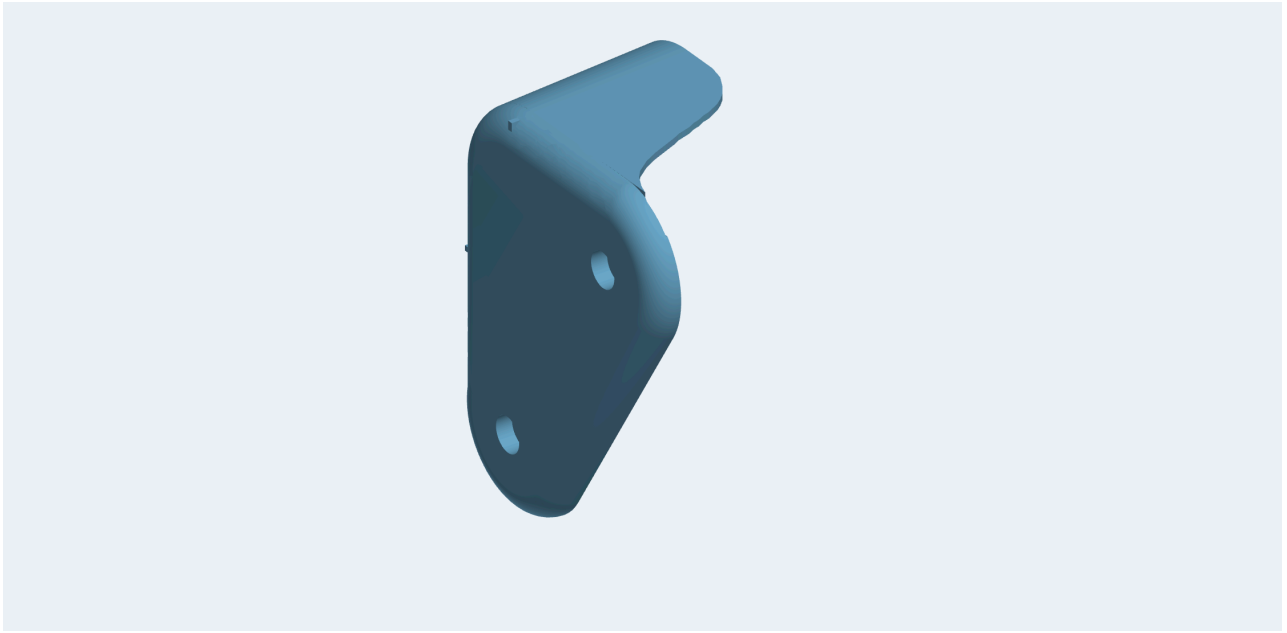


Quantity / 数量: 1

Reference geometry / 参考几何

[STL · 3D view / 查看3D](#)

26 · Upper Leg Left

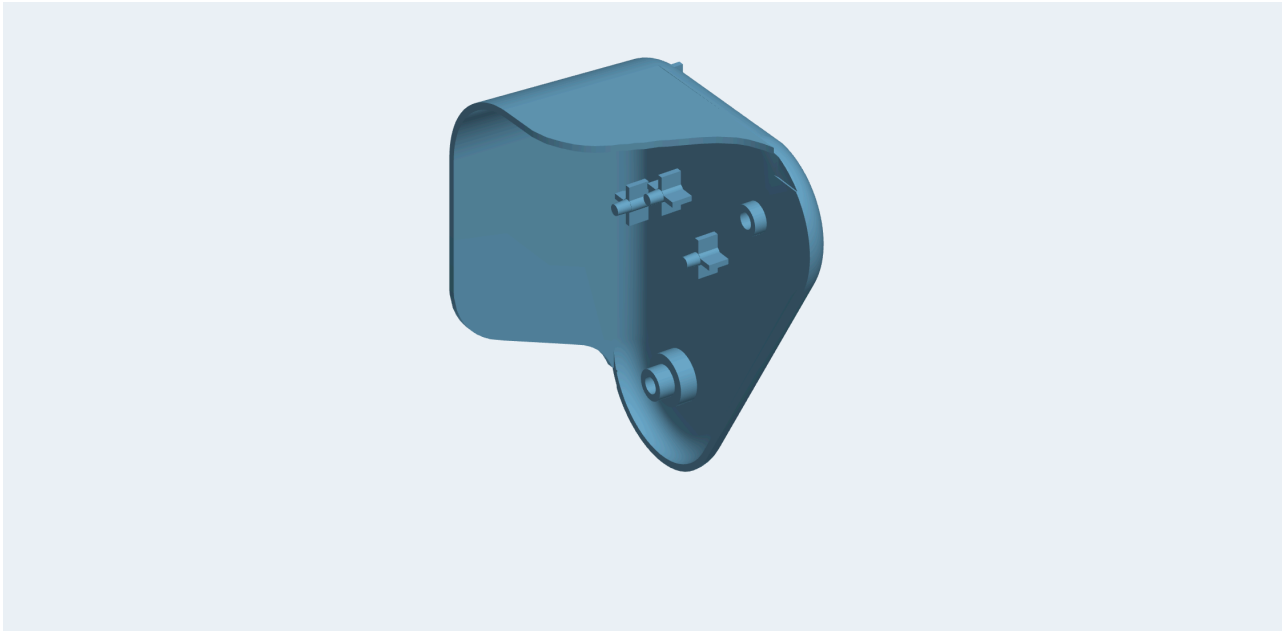


Quantity / 数量: 1

Current design / 当前设计

[STL · 3D view / 查看3D](#)

27 · Upper Leg Right

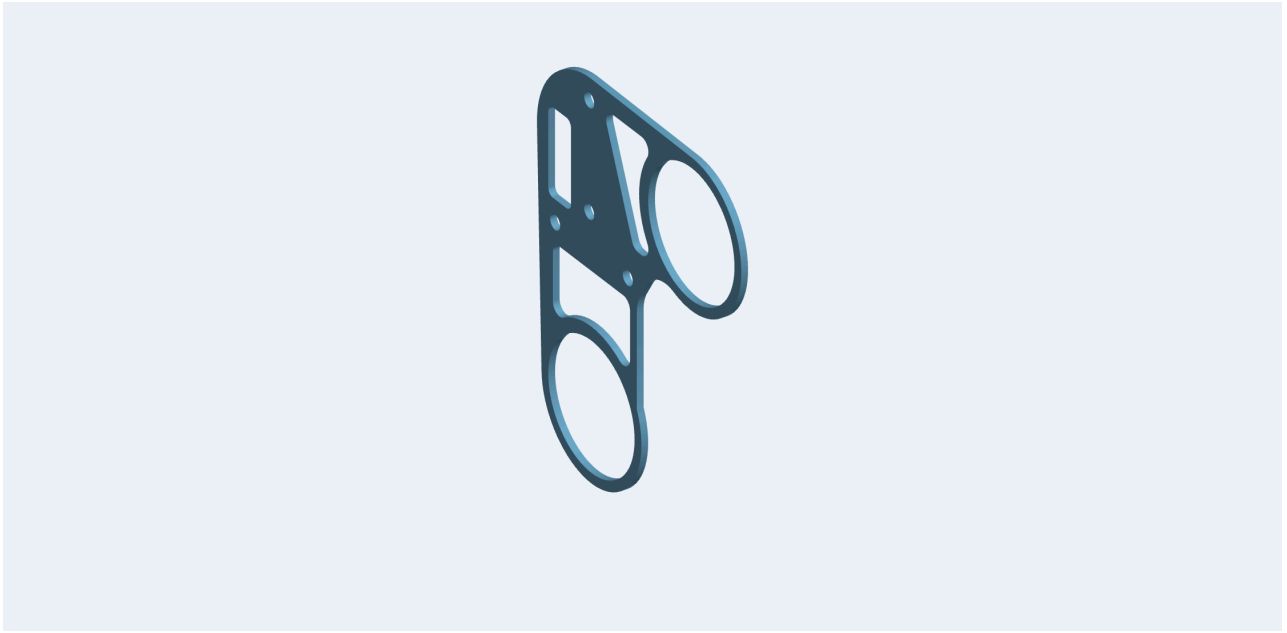


Quantity / 数量: 1

Current design / 当前设计

[STL · 3D view / 查看3D](#)

28 · Upper Leg Rigidity Plate

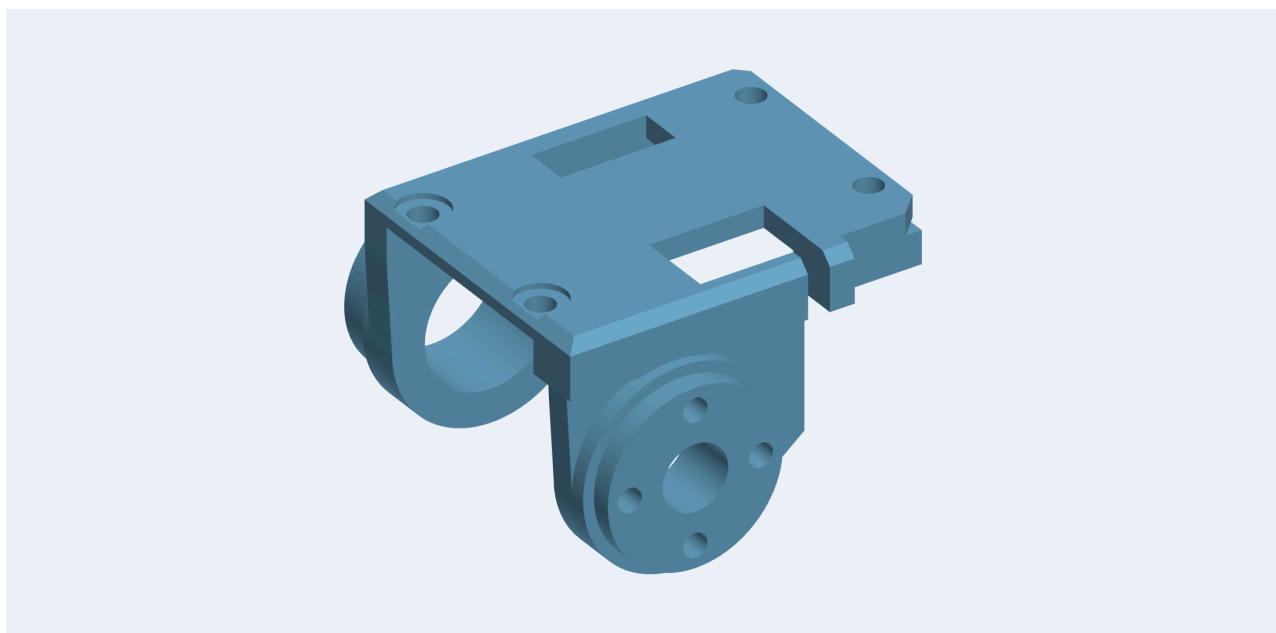


Quantity / 数量: 2

Reference geometry / 参考几何

[STL](#) · [3D view](#) / [查看3D](#)

29 · Yaw Roll Motion

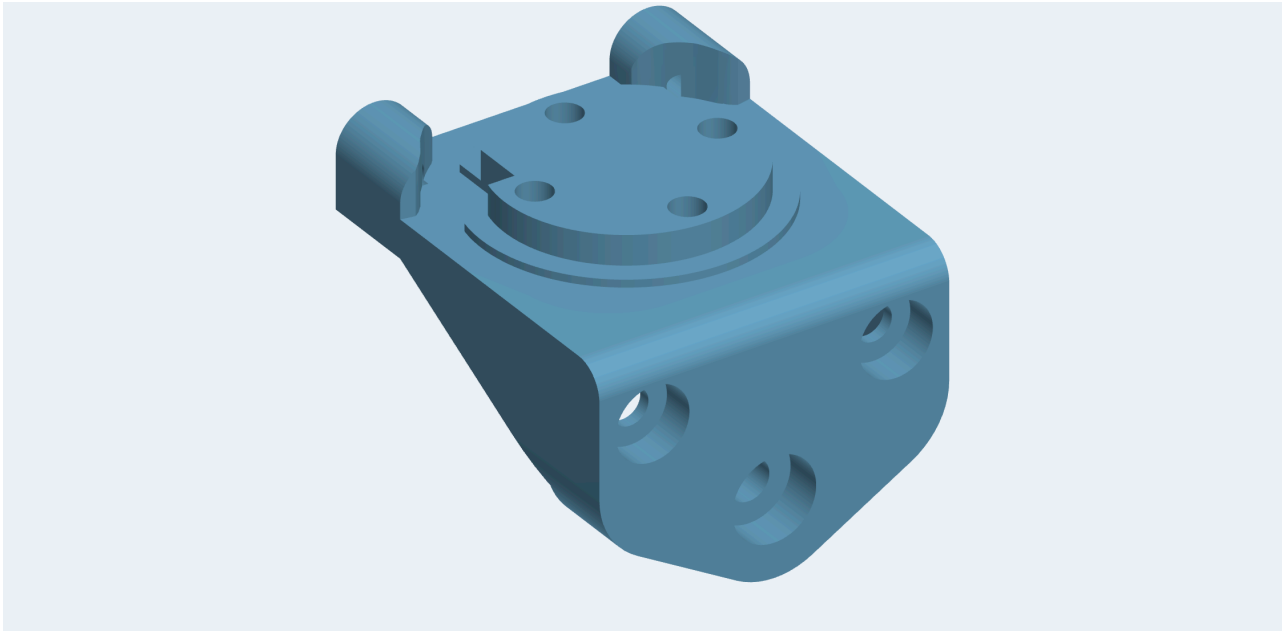


Quantity / 数量: 1

Current design / 当前设计

[STL · 3D view / 查看3D](#)

30 · Yaw2Roll



Quantity / 数量: 2

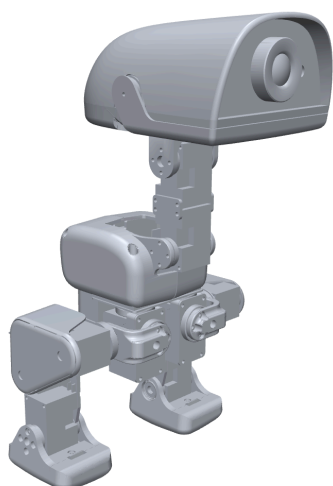
Reconstructed part / 重建零件

[STL · 3D view / 查看3D](#)

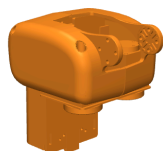
Assembly pictures / 装配示意图

Published reconstruction sequence; physical assembly is not yet verified. / 重建模型装配顺序，尚未经实物验证。

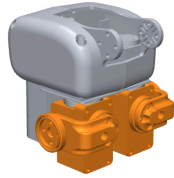
assembled



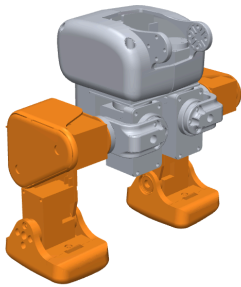
step1



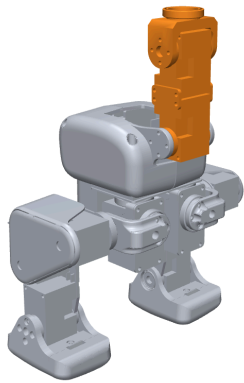
step2



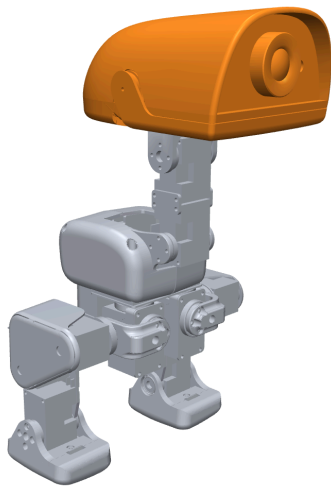
step3



step4



step5



Purchased parts: what changed? / 外购件清单有何变化?

The same32 purchased-part lines remain, with no quantity changes. Three descriptions are clarified: battery/charger kit, unconfirmed ToF module, and unconfirmed speaker specification. The STL list is additional; it does not replace this shopping list.

仍为32项外购件，数量未更改。仅补充电池与充电器套件、待确认的ToF模块及喇叭规格。STL零件清单为新增部分，不替代采购清单。

Original list / 原清单 · Exact changes / 具体变更

ID	Part / 零件	Qty / 数量	Identity / 型号确认
B1	Dynamixel XL330-M288-T smart servo	15	community-derived sub-variant
B2	Robot Cable-X3P 180 mm, 10-pack (Dynamixel bus leads)	2	confirmed at vendor
B3	Radxa ZERO 3W single-board computer	1	FAIL - the specified configuration is absent from the maker's own SKU table
B4	NP-F550-class battery plus one dual charger per robot (whole retail kit)	1	original pack/charger unknown; conditional whole-kit candidates
B6	IMX219 camera board, M12 mount, with lens	1	representative substitute
B7	M12 wide-angle lens (only if the camera board is bought bare)	1	representative substitute; the required focal length is not fixed
B8	22-pin 0.5 mm MIPI CSI FFC ribbon, Radxa to camera	1	class identified; exact part and length unknown
B9	ToF module on four-pin J5/Qwiic interface; original carrier unknown	1	generation varies by unit; module identity and installed geometry unresolved
B10	Ball bearing 22 x 16 x 4 mm (thin section)	11	designation confirmed from dimensions; Pollen's own MPN CANNOT DETERMINE

B1 1	Ball bearing 15 x 10 x 3 mm	3	10 x 15 x 3 mm published mesh envelope confirmed; original installed manufacturer, closure and grade unknown
B1 2	Speaker, reference envelope 35 x 25 x 7 mm (head); original electrical specification unknown	1	unknown original; one priced conditional dimensional candidate, historical oversized offers rejected
B1 3	Microphone(s)	Needs confirmation / 待确认	unknown
B1 4	NFC reader IC + 2 antennas (head and beak)	1	candidate set of two
B1 5	Camera-in-use indicator LED	1	unknown
B1 6	Bluetooth gamepad (Xbox-style)	1	unknown
B1 7	USB-C cable	1	unknown
B1 8a	M2 x 6 socket head cap screw, A2 stainless, DIN 912	80	standard
B1 8 b	M2 socket head cap screws, other lengths (x4, x8, x12) + M2 nuts + M2.5x6	185	standard
B1 8c	M2 heat-set threaded inserts, brass, ~3.0 mm	60	standard class, vendor-specific part
B1 9a	PLA filament (hard printed parts)	0.2182946	matches the sliced preset exactly
B1 9 b	TPU filament (soft parts: soles x2, jaw_soft, soft_mouth_top)	0.0262028	matches the sliced preset exactly
P1 c	TLV320AIC3104IRHBR stereo audio codec (on the Robot HAT)	1	confirmed

P2 c	LSM6DSV16X 6-axis IMU (on the imu_to_dxl board, Dynamixel slave ID 200)	1	confirmed
P1 d	BMI088 6-axis IMU (dormant footprint on the Robot HAT)	1	confirmed
R1	USB-C CC/PD controller fitted on the Radxa ZERO 3W (reference line - NOT separately purchased)	0	two names, one socket - resolved by lane C in favour of ET7301B on the schematic
R 2	MCP73213 2S Li-ion charge controller (in the external charger, not the robot)	0	family confirmed (part:mcp73213 verdict PASS); the exact option suffix a bespoke charger needs depends on the chosen float voltage (8.2 / 8.4 / 8.7 / 8.8 V)
R 3	S-8252 2S battery protection IC (inside the NP-F550 pack, not separately purchased)	0	series confirmed, option unknown
C1	JST EH 3-pin connector kit for the Dynamixel bus, if the harness is crimped rather than bought	1	confirmed - these are exactly the parts wiring/CABLES.md names
C 2	JST SH 4-pin (ToF / Stemma) and GH 2-pin (speaker) housings	1	SH 4-pin confirmed at the HAT's J5; the ToF-board end and the HAT speaker end are CANNOT DETERMINE
P1	Robot HAT - custom PCB, bare board + assembly	1	in-house design, DRC verdict FAIL - quotable as an outline and stackup, not releasable as artwork
P2	imu_to_dxl v2 - custom PCB, bare board + assembly	1	in-house design, DRC verdict FAIL - quotable as an outline and stackup, not releasable as artwork
P 3	Banana battery-contact PCB - custom, bare board + contacts	1	in-house design, DRC verdict CANNOT DETERMINE - the outline and stackup are quotable, the contact part is not

Bought parts shown in the robot / 整机中的外购件

These eight shapes complete the 38 distinct components in the published 70-piece model. Some are simplified envelopes. The shopping list has 32 different procurement lines, including things not represented as separate geometry.

这8种外形补全已发布整机模型的38种零件，部分为简化外形。采购清单仍为32项，包含没有单独几何模型的物料。

31 · Bearing 22X16X4

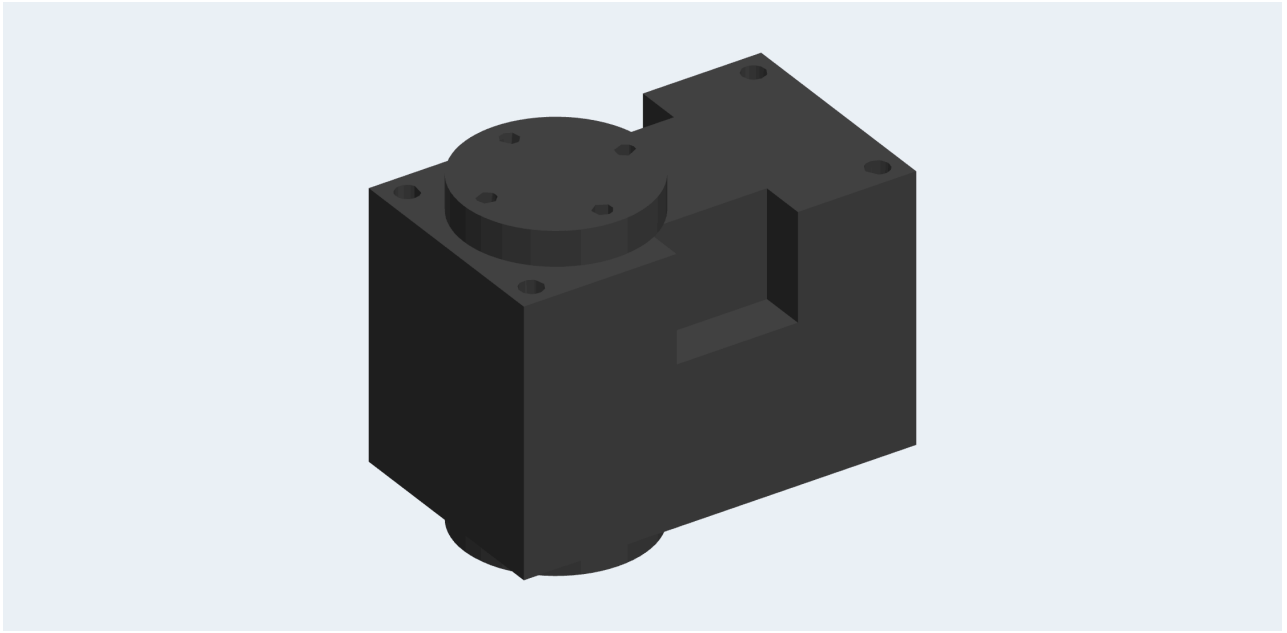


Shown in model / 模型数量: 11

Bought-component representation / 外购件外形，不是打印替代件

[3D view / 查看3D](#) · [STL representation / 模型STL](#)

32 · XI330 M288 T

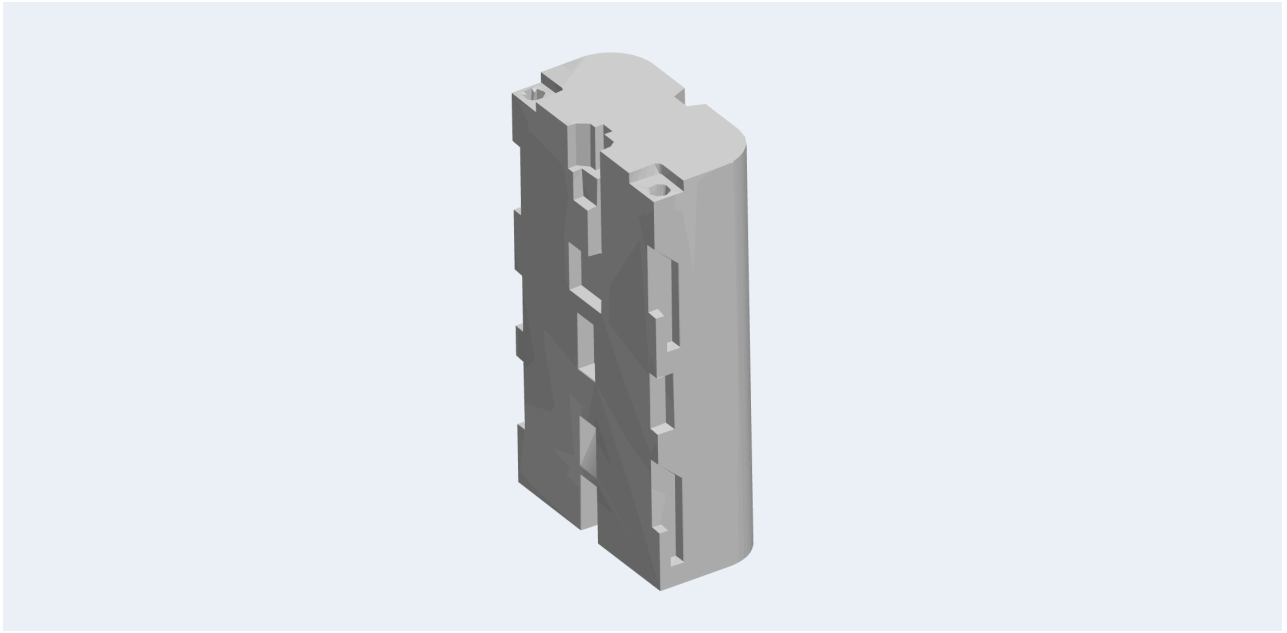


Shown in model / 模型数量: 15

Bought-component representation / 外购件外形，不是打印替代件

[3D view / 查看3D](#) · [STL representation / 模型STL](#)

33 · Np F550

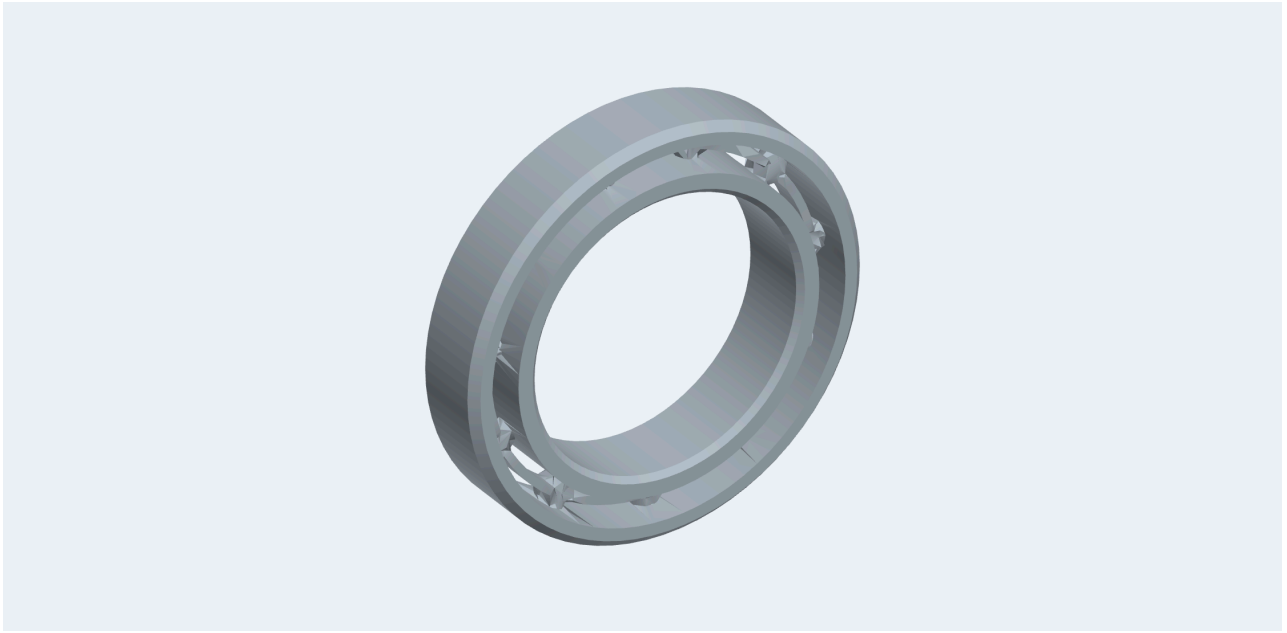


Shown in model / 模型数量: 1

Bought-component representation / 外购件外形，不是打印替代件

[3D view / 查看3D](#) · [STL representation / 模型STL](#)

34 · Bearing 15X10X3

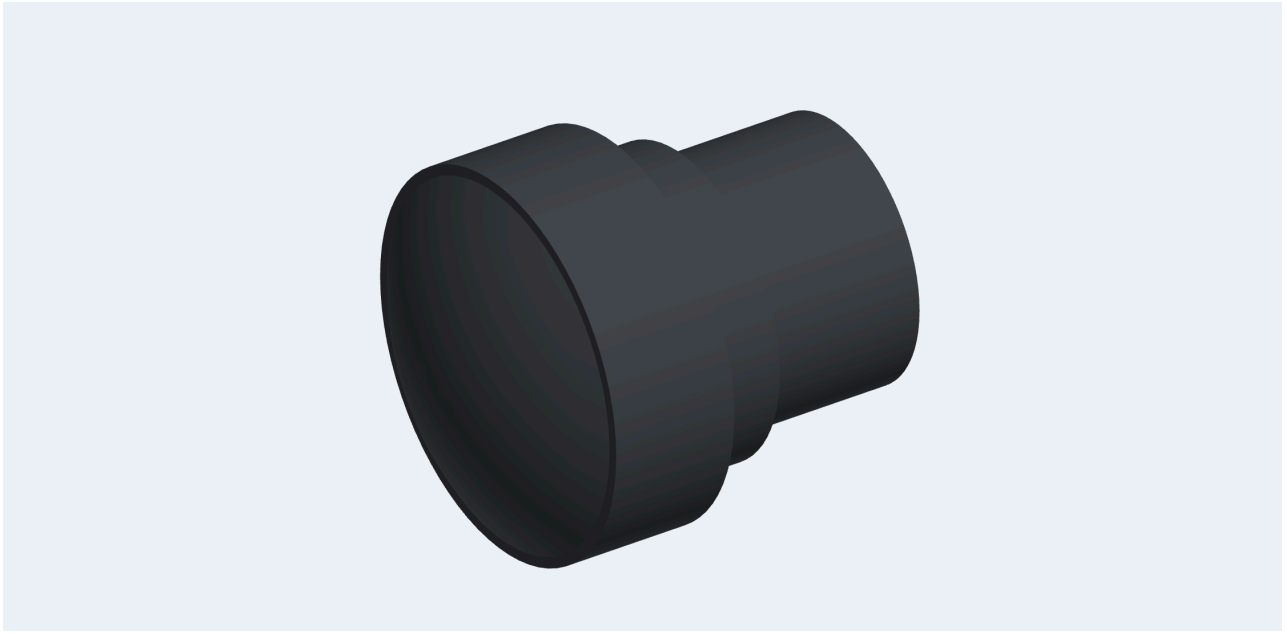


Shown in model / 模型数量: 3

Bought-component representation / 外购件外形，不是打印替代件

[3D view / 查看3D](#) · [STL representation / 模型STL](#)

35 · M12 Lens

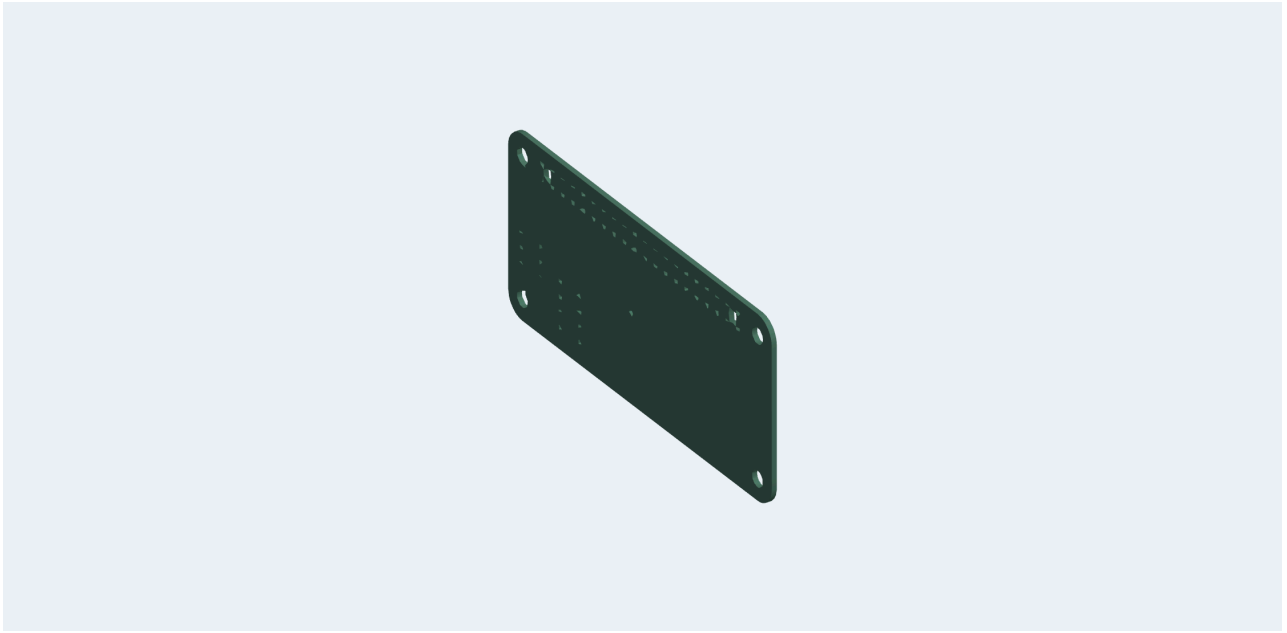


Shown in model / 模型数量: 1

Bought-component representation / 外购件外形，不是打印替代件

[3D view / 查看3D](#) · [STL representation / 模型STL](#)

36 · Robot Hat Pcb



Shown in model / 模型数量: 1

Bought-component representation / 外购件外形，不是打印替代件

[3D view / 查看3D](#) · [STL representation / 模型STL](#)

37 · Radxa Zero 3W

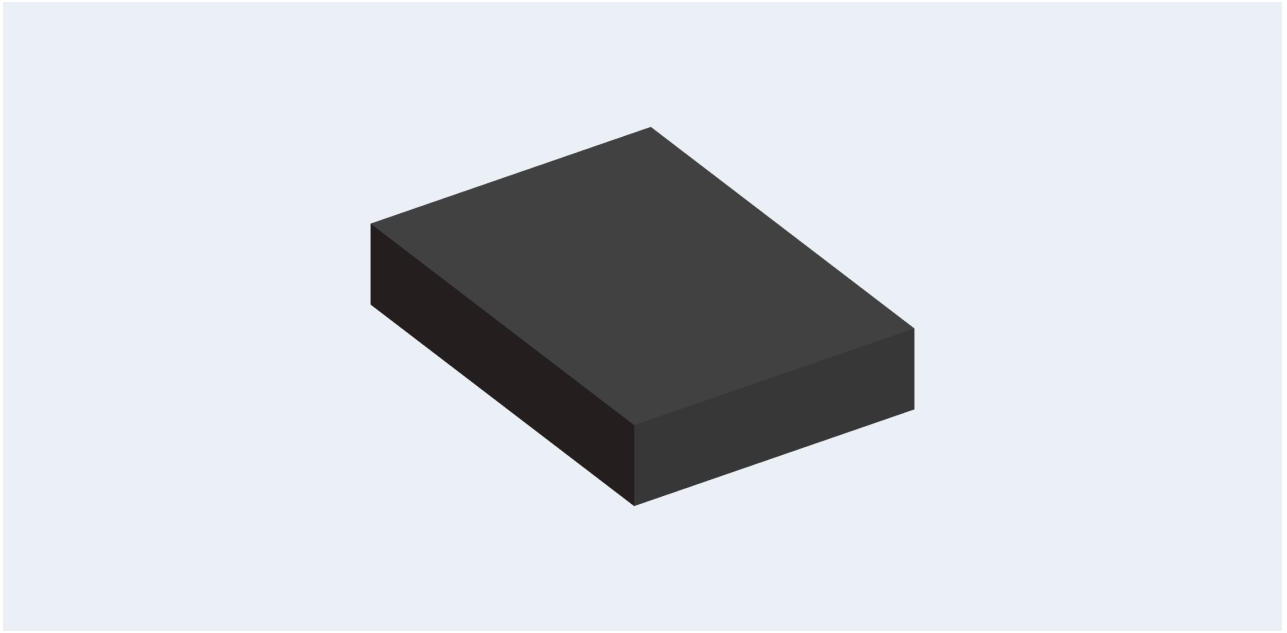


Shown in model / 模型数量: 1

Bought-component representation / 外购件外形，不是打印替代件

[3D view / 查看3D](#) · [STL representation / 模型STL](#)

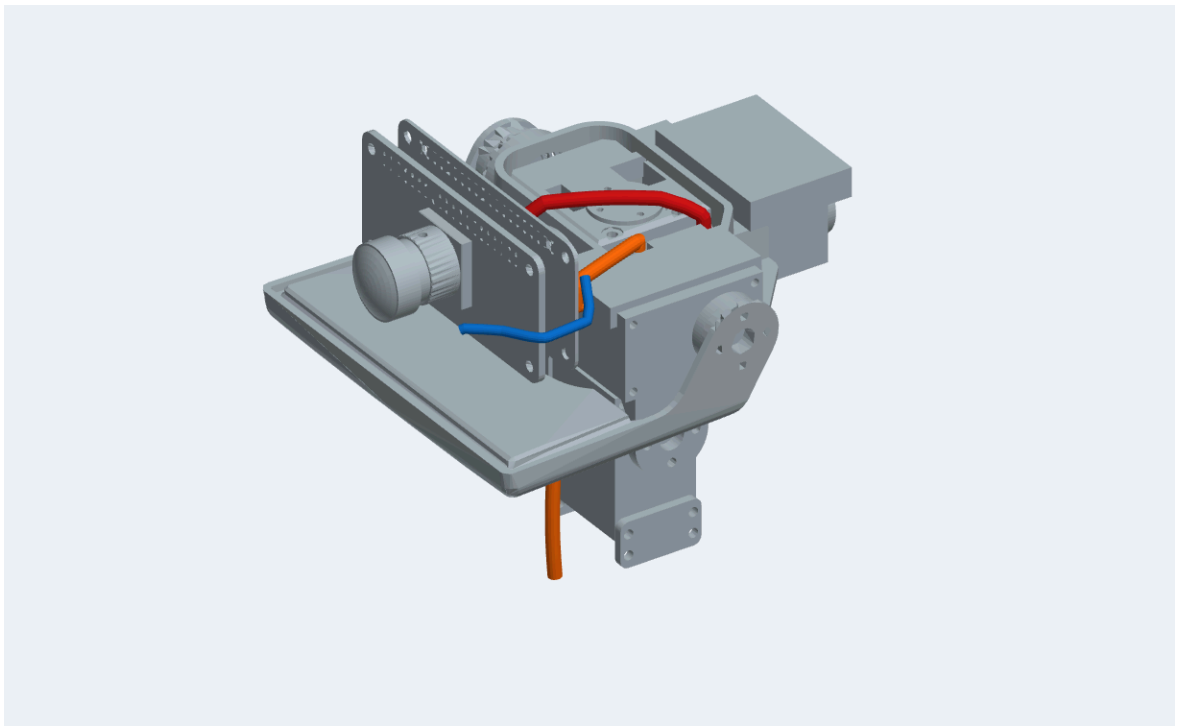
38 · Speaker



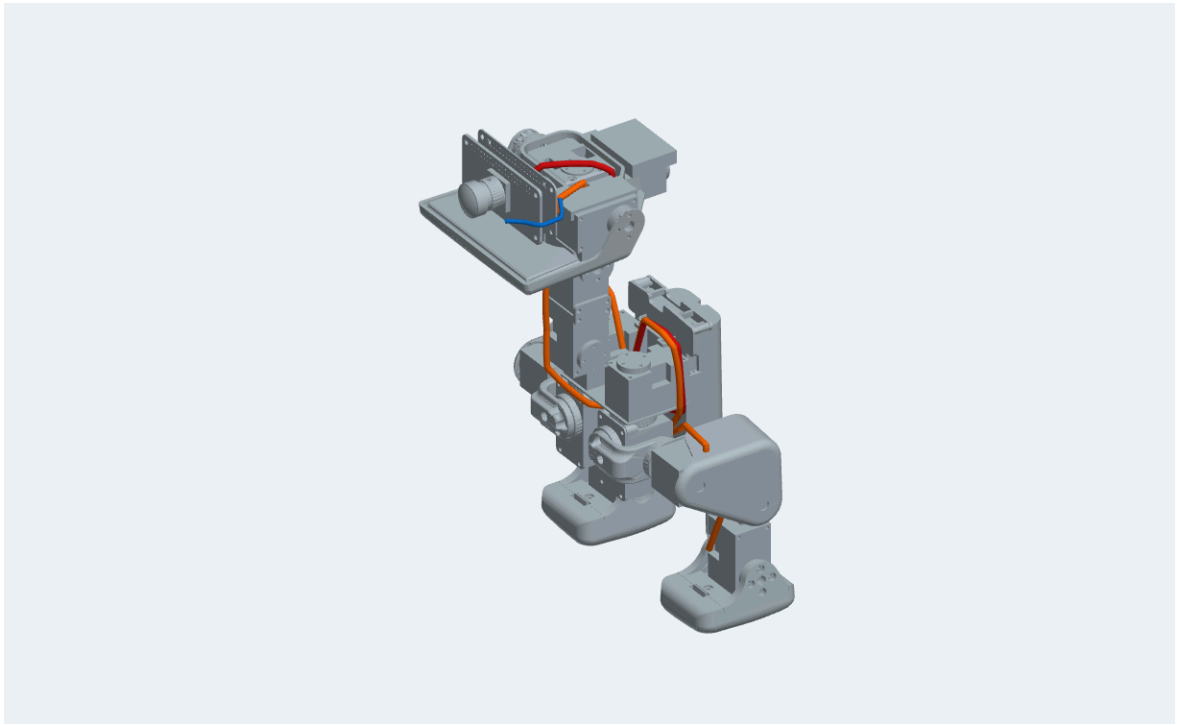
Shown in model / 模型数量: 1

Bought-component representation / 外购件外形，不是打印替代件

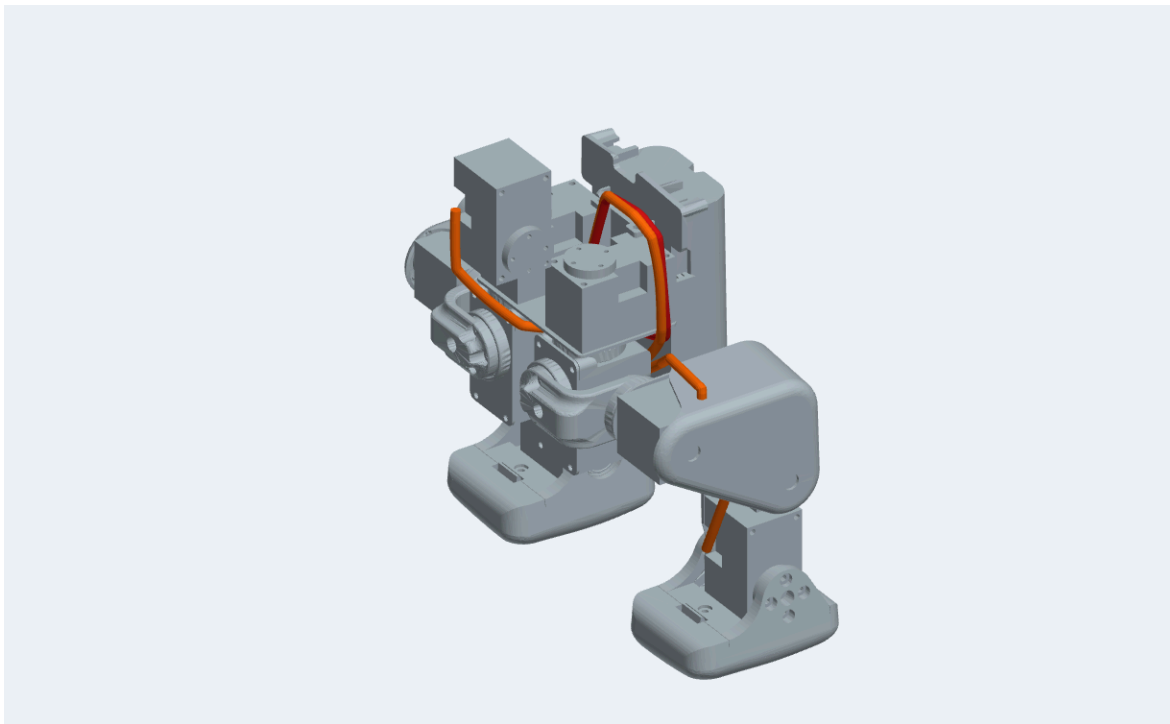
[3D view / 查看3D](#) · [STL representation / 模型STL](#)



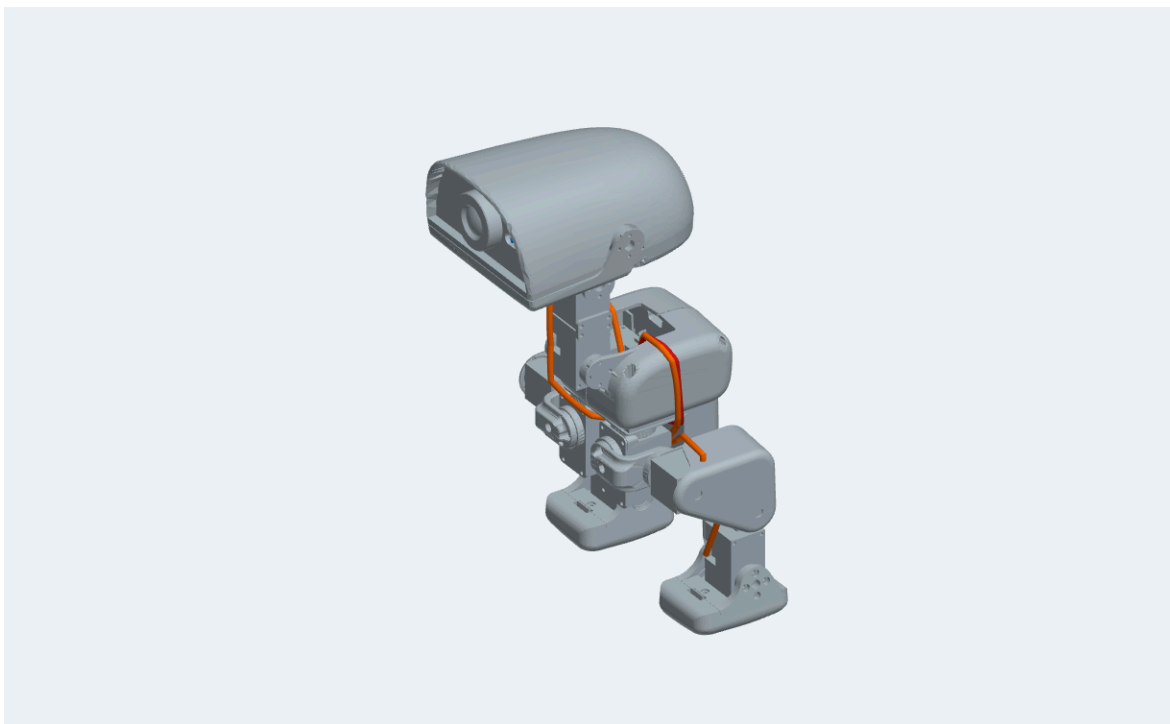
Inside and cables



Leg connections



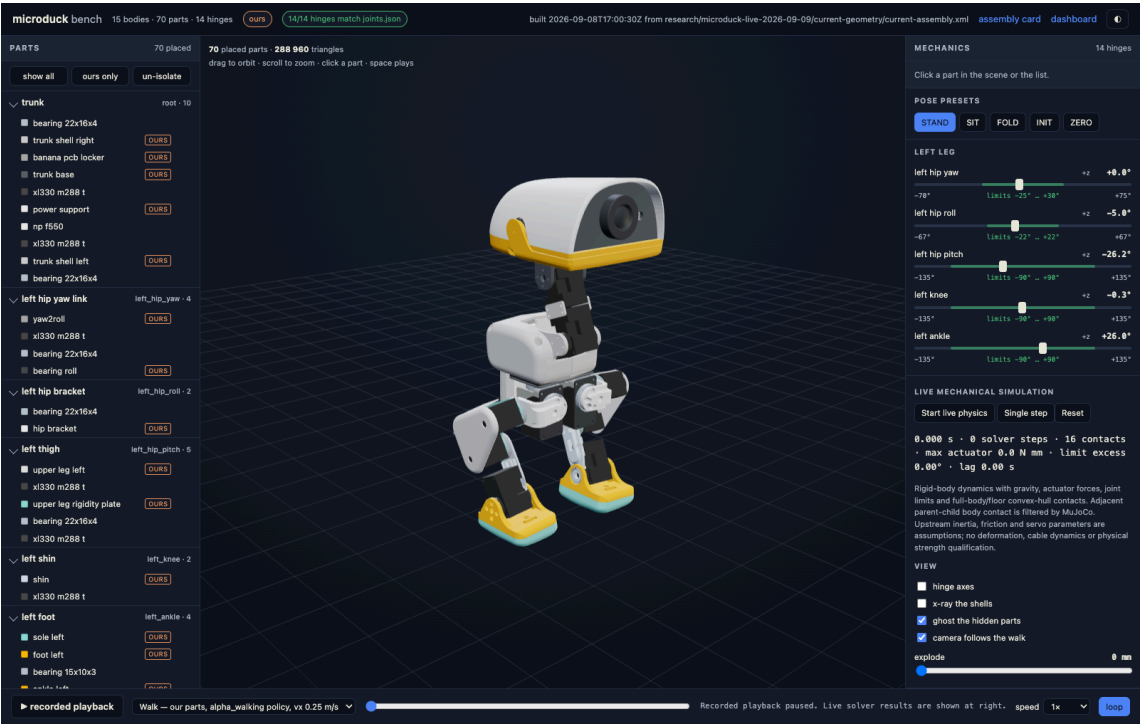
Whole robot and cables



Mechanical simulation / 机械仿真

See the movement and contact tests, and what still needs validation. / 查看运动、接触测试及待验证事项。

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Review one item at a time / 逐项检查

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Inspect each part and connection in 3D, then note its number and your comments. / 逐项查看3D，记录编号及意见。